

Name of the Examination: CAT 2

School	School of Computer Science & Engineering (SCOPE)		
Course Code	CSE 2008	Course Title	Operating Systems
Class Number	AP2021225000204	Date of exam	20/04/2022, FN Session
Slot	C1 + TC1	Faculty Name & ID	Dr. Sanket Mishra, 70217
Duration	1 hr 30 mins	Total Marks	50 Marks

Type your questions here

Q1. For forming a composite compound like carbon-dioxide we require carbon and oxygen. Assume that the elements oxygen and carbon are threads. To form carbon-dioxide we need a mechanism that makes each thread to wait until a complete molecule is ready. As each thread goes through the functionality or mechanism, a bond should be invoked. You need to make sure that all threads from one molecule invoke a bond before any of the threads from the next molecule does.

To simplify the scenario, the following points can be noted:

- If a carbon thread arrives at the mechanism when no other oxygen threads are present, it needs to wait for arrival of two oxygen threads.
- If a oxygen thread arrives at the mechanism when no other threads are found, it needs to wait for one carbon thread and another oxygen thread.
- For formation of a complete molecule, each group should contain a carbon and two oxygen threads.

Write synchronization codes for carbon and oxygen that implements the above conditions.

You need to use the following variables to formulate the synchronization code.

- `mutex = semaphore(1)`
- `carbon = oxygen = 0` (initially)
- `mechanism = mechanism(3)` (since 3 threads form one complete group)
- `oxygenQueue = carbonQueue = semaphore(0)`

oxygen and carbon can be taken as counter variables and need to be protected using mutex. Mechanism is the part where each set of all three threads (2 oxygen and 1 carbon) meet while initiating bond while allowing next set of threads to proceed. oxygenQueue maintains semaphore of oxygen threads that are waiting while carbonQueue maintains the list of carbon threads that are

waiting. Wait and signal operations can be used with the queues to initiate the intake and release of threads.

Q2. In a system snapshot, it was found that two processes P_x and P_y are executing concurrently. They have a shared resource which is a variable "sum" with an initial value of 0. However, mutual exclusion is not imposed while the processes access this value. Each of the process uses a loop that iterates for 50 times and contains a single line that increments the sum value by 1. If no other operations are performed on the sum identify what will be the probable maximum and minimum values of the sum when both P_x and P_y terminate.

Q3. Consider the following system with five processes <P₁, P₂, P₃, P₄, P₅> and four resources <R₁, R₂, R₃, R₄>. The following table shows the snapshot of the system at time = 0. [12.5]

Process	Allocated				Maximum				Available			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
P1	0	0	1	2	1	4	2	2	1	5	2	0
P2	1	0	2	1	1	9	7	6				
P3	1	0	5	4	2	5	9	6				
P4	0	6	3	2	0	6	5	3				
P5	1	0	1	4	1	7	6	7				

Determine the state is safe or not using Banker's algorithm and show safe sequence. If P₂ requests <0,4,2,0> resources can it be granted?

Q4. The following partitions of memory are provided. All the partitions are free or unallocated except the OS partition which is occupied by Windows. The goal of the OS designer is to identify the schemes which minimizes internal fragmentation. Which scheme(s) should be avoided that maximizes internal fragmentation during process allocation? Name the schemes (if any) that also exhibit external fragmentation.

OS	100k	20k	80k	30k	40k
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QR Mapping

[illegible]

Name of the Examination: CAT 2

School	School of Computer Science & Engineering (SCOPE)		
Course Code	CSE 2008	Course Title	Operating Systems
Class Number	AP2021225000205	Date of exam	20/04/2022, AN Session
Slot	C2 + TC2	Faculty Name & ID	Dr. Sanket Mishra, 70217
Duration	1 hr 30 mins	Total Marks	50 Marks

Type your questions here

Q1. Consider the following system with five processes <P1, P2, P3, P4, P5> and three resources <R1, R2, R3>. The following table shows the snapshot of the system at time = 0.

Process	Allocated			Maximum			Available		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P1	1	1	2	4	3	2	3	1	0
P2	2	1	2	3	2	2			
P3	4	0	1	9	2	2			
P4	0	2	0	7	6	0			
P5	1	1	2	8	2	2			

Determine the state is safe or not using Banker's algorithm. Also state the safe sequence.

Q2. Study the code segment and answer the question.

```

void diningPhilosophers()
{
    while (true)
    {
        Think();
        Eat();
    }
}

```

```

void eat()
{
    pickupLeftFork();
    pickupRightFork();
    eatA While();
    putDownRightFork();
    putDownLeftFork();
}

```

Does the eat () in the above code prevent starvation ? If the statements pickupLeftFork() and pickupRightFork() are replaced with pickupBothForks(), will it prevent deadlock and promote starvation ? Give valuable reasons in support of your answer.

Q3. Suppose there is a “merry go round” ride in Wonderla Amusement Park containing “x” passenger threads and a car thread. Passengers who have procured tickets wait to take their seats in the ride that can hold “N” passengers where $N < x$. The ride can start movement on the track once it is occupied fully.

Some constraints to be considered in this problem are as follows :

- Passengers can invoke board() and unboard() for getting in and out of the ride.
- The passenger car can initiate load(), run() and unload() functionality.
- Passengers cannot get into the ride until car has initiated load().
- Car cannot leave unless N passengers have boarded.
- Passengers cannot alight or get off the ride unless car has initiated unload().

Write a pseudocode for the passengers and car that enforces these conditions. You need to use the following variables while writing the code.

- mux = semaphore(1) for boarders and mux2 = semaphore(1) for unboarders.
- boarders = 0, unboarders = 0 (initially).
- boardQueue = semaphore(0), unboardQueue = semaphore(0)
- allAboard = AllAshore = semaphore(0).

mux protects passengers by facilitating mutual exclusion. It also identifies the count of passengers that have invoked boardCar. Passengers maintain boardQueue before boarding where they wait for their turn and unboardQueue before unboarding. allAboard indicates that the car is full and allAshore means car is empty.

Q4. In the given diagram the address space is provided where the shaded portions depict occupied portions and unshaded parts depict the holes. The unit of measurement is Kilobytes. Allocate the

processes given in the table in the holes using a policy that exhibits a slower performance in every iteration for searching the most suitable hole. Implement another memory allocation scheme that is fast and does not use any underlying idea in allocation, however, is repetitive in nature as it starts from the beginning in every iteration.



Process	Size in KBs
P1	15
P2	50
P3	90
P4	75
P5	120
P6	10

QP Mapping

[illegible]

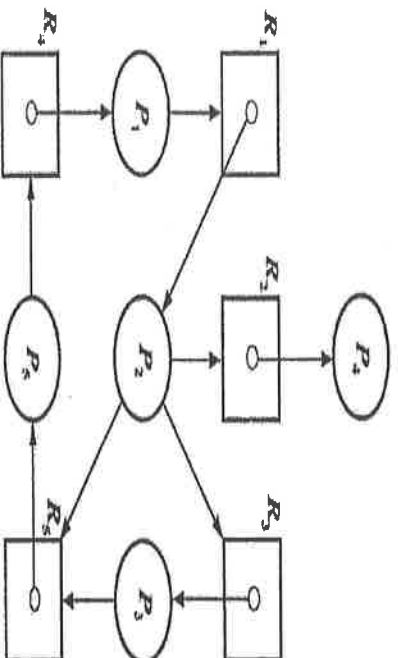
Name of the Examination: CAT 2

School	School of Computer Science and Engineering		
Course Code	CSE2008	Course Title	Operating System
Class Number	AP2021225000206	Date of exam	20/04/2022
Slot	C1+TC1	Faculty Name & ID	Dr. Sushil Kumar Tiwari (70287)
Duration	1 hour 30 Minutes	Total Marks	50

1. Suppose, there are multiple users who are accessing the IRCTC database. Some users are accessing the database to check the seat availability and the PNR status. Other users are accessing the same database to book the tickets. How the synchronization is involved in the above discussed scenario? What problems may occur if the synchronization is not ensured among these users. Also, justify your answer. [12.5]

2. Consider the given resource allocation graph. Do the followings: [12.5]

- (i) Convert it to the matrix representation (i.e., Allocation, Request, Available).
- (ii) Is there a deadlock? if there is a deadlock, which processes are involved?



3. Why non-contiguous memory is preferable instead of contiguous memory allocation. Is it possible to remove the entire memory wastage from main memory using non-contiguous memory allocation scheme? If not, then which type of memory wastage is still involved in non-contiguous allocation method. Is there a guarantee to utilize the main memory more efficiently by decreasing the size of page? Justify your answer. [12.5]

4. Consider a simple paging system. It has 2^{22} Bytes physical memory and page size of 512 Bytes. The memory is Byte addressable. Then, what is approximate page table size if page table entry is 4 Bytes. Also, determine the number frames available in main memory. [12.5]

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	PO3, PO5			12.5
Q2	3	CO7	PO4, PO10			12.5
Q3	4	CO5	PO7, PO10			12.5
Q4	4	CO5	PO7, PO10			12.5

Name of the Examination: CAT 1

School	School of Computer Science and Engineering			
Course Code	CSE2008	Course Title	Operating System	
Class Number	AP2021225000207	Date of exam	20/04/2022	
Slot	C2+TC2	Faculty Name & ID	Dr. Sushil Kumar Tiwari (70287)	
Duration	1 hour 30 Minutes	Total Marks	50	

1. Suppose, you have to implement the mechanism to prevent the system from deadlock. What mechanism will you apply so that circular wait does not exist. If mechanism succeeded to prevent the system from the deadlock. Then, what limitations might be involved in order to get the system prevented from the deadlock? [12.5]

2. (a) Consider, a system is having three processes which requires a certain unit of resource R as listed in given table. What should be the minimum number of units of R, so that deadlock will not occur. Also, justify your answer. [6.5]

Process-ID	Units of resource R
P1	12
P2	32
P3	63

(b) If there are 6 units of resource R in the system and each process in the system requires 3 units of resource R, then how many processes in the system can be present at maximum so that no deadlock will occur? [6]

3. (a) Consider six memory partitions of size 600 KB, 500 KB, 200 KB, 400 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 46 KB, 357 KB, 210 KB and 489 KB in that order. Perform the allocation of processes using First fit, best fit and worst fit policies. [6.5]

(b) Consider a system with TLB access time 10 ns and main memory access time 50 ns. Calculate effective memory access time if TLB hit ratio is 55%. [6]

4. Suppose, there are two computer systems. One is using the fixed size partition and other is using the variable size partitions method. Then, which type of fragmentations are involved in these systems. How the degree of multiprogramming is limited in the fixed size partitions. Is there always a guarantee to have a higher degree of multiprogramming in variable size partition in comparison to Fixed size partition. Justify, your answer. [12.5]

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	PO3, PO5			12.5
Q2	3	CO7	PO4, PO10			12.5
Q3	4	CO5	PO7, PO10			12.5
Q4	4	CO5	PO7, PO10			12.5

Name of the Examination: CAT 2

School	School of Computer Science and Engineering			
Course Code	CSE2008	Course Title	Operating System	
Class Number	AP2021225000208	Date of exam	22/04/2022	
Slot	E1+TE1	Faculty Name & ID	Dr. Sushil Kumar Tiwari (70287)	
Duration	1 hour 30 Minutes	Total Marks	50	

1. A single processor system has three resource types X, Y and Z, which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the column alloc denotes the number of units of each resource type allocated to each process, and the column request denotes the number of units of each resource type requested by a process in order to complete execution. Determine the safe sequence of the processes so that deadlock can be avoided. [12.5]

	Alloc			Request		
	X	Y	Z	X	Y	Z
P0	1	2	1	1	0	3
P1	2	0	1	0	1	2
P2	2	2	1	1	2	0

2. Consider two cooperative processes P1 and P2 are accessing their critical sections whenever needed. Write any pseudocode for processes P1 and P2 which ensure the mutual exclusion between them. Also, discuss what are the other necessary conditions which must exist in order to ensure the synchronization between multiple cooperative processes. [12.5]

3. In Paging, CPU requires to access the main memory twice to get the requested page. Suggest the approach which can be utilize to reduce the memory access time. In addition, identify the type of memory fragmentation which is involved with paging. Is it possible to remove memory wastage completely? If no, then justify your answer. [12.5]

4. Suppose, a computer system uses the paging mechanism for the main memory allocation. It considers the page size of 8 KB and a physical address of 32 bits. Here, page table entry contains some optional fields apart from the frame number. The page table uses the 5 bits to store the optional fields. If the maximum size of the page table of a process is 24 megabytes. Then, determine the number bits required to represent the logical address. [12.5]

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	PO3, PO5			12.5
Q2	3	CO7	PO4, PO10			12.5
Q3	4	CO5	PO7, PO10			12.5
Q4	4	CO5	PO7, PO10			12.5

QUESTION PAPER

Name of the Examination: CAT ~ 2 (Winter 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: C1+TC1

Date of Exam: 20-04-2022

Duration: 90 min

Total Marks: 50

Class ID: AP2021225000209

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. A schedule contains two concurrent processes P0, P1 that share two common variables Q, M with initial values 2, 3 respectively. P0 updates Q and M using two instructions S10 and S11, P1 updates the variables Q, M using S20 and S21 instructions. What would be the different values that Q, M can possibly take after the execution of the processes concurrently? **(12.5M)**

Process P0:	Process P1:
S10: $Q=Q+M$	S20: $Q=Q-1$;
S11: $M=M+Q$;	S21: $M=M+1$;

Q2. a) A computer system has 12 tape drives, which are shared by three concurrent processes P1, P2, and P3. Currently, P1, P2, and P3 are allocated with 5, 2, and 3 tape drives respectively. The maximum requirements of P1, P2, and P3 are 8, 5, and 5 tape drives respectively. Then, can you guarantee that the system is in safe state? If so, then what will be the safe sequence? **(6M)**

b) Consider a computer system that has N numbers of magnetic tape drives, which are shared among three concurrent processes P1, P2, and P3. The individual maximum demands of P1, P2, and P3 are 4, 6, and 9 respectively. It is added that the combined maximum demand of P1, and P3 together will never exceed 10. In this scenario, what will be the MINIMUM value of N that can ensure the system to be deadlock free? Explain your answer. **(6.5M)**

Q3. a) Given five memory partitions of 250 KB, 400 KB, 360 KB, 500 KB, and 180 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of 315 KB, 221 KB, 135 KB, and 284 KB (in order)? Which algorithm makes the most efficient use of memory? **(8.5M)**

b) Consider the following segment table:

Segment	Base	Length
0	620	450
1	1800	60
2	170	74
3	2145	321
4	1450	113

What are the physical addresses for the following logical addresses? (4M)

- 0,512
- 1,47
- 2,13
- 3,185

Q4. Consider a process with 7 pages. If there are 3 frames in main memory, and the reference string is 2, 4, 1, 7, 1, 4, 2, 6, 1, 6, 2, 1, 6, 2, 3, 2, 6, 1, 5, 1, then how many page faults would occur with the following replacement algorithms?

- LRU replacement
- Optimal replacement

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	PO1, PO2, PO3, PO5	PEO1, PEO3	PSO1, PSO2, PSO4	12.5
Q2	3	CO7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	12.5
Q3	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3, PSO4	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3, PSO4	12.5

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: C2+TC2

Date of Exam: 20-04-2022

Duration: 90 min

Total Marks: 50

Class ID: AP2021225000210

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Consider that there is a central database which can be accessed by two types of processes called Readers and Writers. A Reader process reads data from the database and a Writer process writes data to the database. Any number of readers can read data from the database at a time. When a writer is writing data to the database, no other writer and reader is allowed to access the database. Write monitor solution to provide synchronous access to the database. (12.5M)

Q2. Consider the following snapshot of a system (A, B, C, D are the resources, and P0 to P4 are the processes):

	Max	Allocation	Available
	A B C D	A B C D	A B C D
P0	2 2 0 1	2 1 0 1	1 2 0 1
P1	2 6 4 2	0 0 0 1	
P2	3 4 5 3	2 2 5 1	
P3	5 5 4 1	0 4 2 1	
P4	0 4 5 3	0 2 1 0	

Answer the following questions using the Banker's algorithm:

- a. What is the content of the matrix Need?
- b. Is the system in a safe state? If so, write the safe sequence.
- c. If the request from process P2 arrives for (0, 2, 0, 0) can the request be granted? What will be the state of the system then? (12.5M)

Q3. a) Assume the memory is divided into six partitions with size 200 KB, 400 KB, 100 KB, 600 KB, 300 KB, 500 KB (in order). Calculate the total internal fragmentation when the processes with size 336 KB, 96 KB, 173 KB, 415 KB, 245 KB (in order) are placed with i) first-fit algorithm ii) best-fit algorithm. (8.5M)

- b) Consider a logical address space of 256 pages with 64 words per page mapped onto a physical memory of 4096 frames. i) How many bits are required in the logical address? ii) How many bits are required in the physical address? (4M)

Q4. Consider these two policies of page replacement where policy A replaces the page which is referred more number of times in the past and policy B replaces the page which is referred less number of times. If there are 3 frames in main memory, and the reference string is 1 1 2 1 2 1 3 2 7 8 4 0 1 2, then how many page faults will occur for both policies A and B. **(12.5M)**

QP Mapping

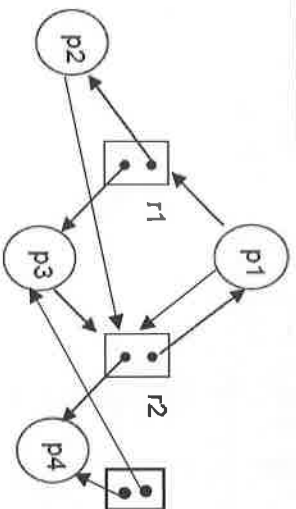
Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	PO1, PO2, PO3, PO5	PEO1, PEO3	PSO1, PSO2, PSO4	12.5
Q2	3	CO7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	12.5
Q3	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3, PSO4	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3, PSO4	12.5

Name of the Examination: CAT 2

SCOPE			
School			
Course Code	CSE2008	Name of the Course	Operating Systems
Semester	WIN SEM (2021-22)	Date of exam	23.04.2022
Slot	F2+TF2	Faculty Name	Dr.S.P. Siddique Ibrahim
Time	2:00 PM to 3:30 PM	Total Marks	50
Class ID	AP2021225000211		

Question No.	Question	Module Number	CO Mapping	Marks																																																								
Q1 a.	There are three resources and five processes in a system. Availability Table: <table><tr><th>Resources</th><th>Quantity</th></tr><tr><td>A</td><td>2</td></tr><tr><td>B</td><td>2</td></tr><tr><td>C</td><td>1</td></tr></table> The quantities of resources requested by the processes and resources allocated by the system are as follows: Request Table: <table><tr><th>Process</th><th>Resource A</th><th>Resource B</th><th>Resource C</th></tr><tr><td>P1</td><td>2</td><td>1</td><td>2</td></tr><tr><td>P2</td><td>3</td><td>2</td><td>2</td></tr><tr><td>P3</td><td>1</td><td>2</td><td>1</td></tr><tr><td>P4</td><td>2</td><td>2</td><td>1</td></tr><tr><td>P5</td><td>3</td><td>4</td><td>0</td></tr></table> Allocation Table: <table><tr><th>Process</th><th>Resource A</th><th>Resource B</th><th>Resource C</th></tr><tr><td>P1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>P2</td><td>2</td><td>0</td><td>1</td></tr><tr><td>P3</td><td>0</td><td>1</td><td>0</td></tr><tr><td>P4</td><td>1</td><td>1</td><td>1</td></tr><tr><td>P5</td><td>2</td><td>0</td><td>1</td></tr></table>	Resources	Quantity	A	2	B	2	C	1	Process	Resource A	Resource B	Resource C	P1	2	1	2	P2	3	2	2	P3	1	2	1	P4	2	2	1	P5	3	4	0	Process	Resource A	Resource B	Resource C	P1	1	1	0	P2	2	0	1	P3	0	1	0	P4	1	1	1	P5	2	0	1			
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P5	2	0	1																																																									
		Module 3	CO7	7.5 Marks																																																								

	i) Verify the system in a deadlock condition? If it so, then what are the process(es) that need to be preempted in order to overcome the deadlock?																							
Q1 b.	Consider there are seven researchers went to hotel and sitting around the circular table. The dining table has seven forks kept beside the researchers and a bowl of fried rice is placed on the center, when any one wants to eat, he/she uses two forks must be shared from neighbour researchers (One from left and another from the right). When a researchers want to think, he/she keeps down both the forks on the table. The researchers are names as R1,R2,R3,R4,R5,R6 and R7. i) How many researchers can get a chance at a same time for eating? ii) If R3 and R4 are tried to eat, who will get the chance first?	Module 3	CO3	5 Marks																				
Q2 a.	How to improve the system performance with respect to the following issues: i. Internal fragmentation ii. External fragmentation	Module 4	CO5	2.5 Marks																				
Q2 b.	Consider the following primary memory holes are available while scanning 100 KB, 350 KB, 450 KB, 600 KB, 700 KB, 400 KB, 300KB, 250KB and 80 KB. These partitions need to be allocated to nine processes, all the processes are arriving at 0 ms. The process requirement is given below: <table><tr><th>P. No</th><th>Memory Requirement</th></tr><tr><td>100</td><td>92KB</td></tr><tr><td>110</td><td>170KB</td></tr><tr><td>120</td><td>380KB</td></tr><tr><td>160</td><td>410KB</td></tr><tr><td>175</td><td>60KB</td></tr><tr><td>190</td><td>155KB</td></tr><tr><td>200</td><td>210KB</td></tr><tr><td>230</td><td>650KB</td></tr><tr><td>280</td><td>120KB</td></tr></table>	P. No	Memory Requirement	100	92KB	110	170KB	120	380KB	160	410KB	175	60KB	190	155KB	200	210KB	230	650KB	280	120KB	Module 4	CO5	10 Marks
P. No	Memory Requirement																							
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175	60KB																							
190	155KB																							
200	210KB																							
230	650KB																							
280	120KB																							
Q3 a.	Perform the allocation of processes using- 1. First Fit Algorithm 2. Best Fit Algorithm 3. Worst Fit Algorithm Consider the following Resource Allocation graph. Check is there any deadlock present, how many cycles in the given allocation. If so, which processes are involved in deadlock?	Module 3	CO7	10 Marks																				



Q3 b. List any five real world situations of deadlock that are not relation with computer system environment. Module 3 CO7 2.5 Marks

Q4 Assume that there are five processes, P_0 through P_4 , and four types of resources. At T_0 we have the following system state:

P. No	Allocation				Max Demand			
	A	B	C	D	A	B	C	D
P0	4	0	0	1	6	0	1	2
P1	1	1	0	0	1	7	5	0
P2	1	2	5	4	2	3	5	6
P3	0	6	3	3	1	6	5	3
P4	0	2	1	2	1	6	5	6

Module 3 CO7 12.5 Marks

- i) Using Banker's algorithm, illustrate that the system is in safe state by demonstrating an order in which the processes may complete.
 - a. Available (3,2,1,1)
- ii) If a request from process P_4 arrives for additional resources of (0,1,0,0), can the Banker's algorithm grant the request immediately? Present the new system state.

Q. No.	Module Number	CO Mapped	PO Mapped	PSO Mapped	PEO Mapped	Marks
Q1 a	Module: 3	CO7	PO2, PO3, PO4,	PO2, PO3, PO4,	PO2, PO3, PO4,	7.5
Q1 b	Module: 3	CO3	PO1, PO2, PO3, PO5	PSO1, PSO2, PSO4	PEO1, PEO3	5
Q2.a	Module: 4	CO5	PO2, PO3, PO4, PO7,	PO2, PO3, PO4, PO7,	PO2, PO3, PO4, PO7,	2.5
Q2.b	Module: 4	CO5	PO2, PO3, PO4, PO7,	PO2, PO3, PO4, PO7,	PO2, PO3, PO4, PO7,	10
Q3 a	Module: 3	CO7	PO2, PO3, PO4,	PO2, PO3, PO4,	PO2, PO3, PO4,	10
Q3 b	Module: 3	CO7	PO2, PO3, PO4,	PO2, PO3, PO4,	PO2, PO3, PO4,	2.5

Q4	Module: 3	CO7	PO2, PO3, PO4,	PO2, PO3, PO4,	PO2, PO3, PO4,	12.5
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Name of the Examination: CAT 2

School	SCOPE		
Course Code	CSE2008	Course Title	Operating Systems
Class ID	AP2021225000212		
Semester	WIN 2021-22	Date of exam	20.4.2022
Slot	C1+TC1	Faculty Name & ID	Dr. Sanjit Ningthoujam, 70147
Duration	90 mins	Total Marks	50

Type your questions here

Q1. A) Consider the system with three processes P1, P2, and P3. The maximum demand of each process is 5, 9 and 13 respectively for the resource R. What is the minimum no. of resource require to ensure deadlock free operation? (6.5 marks)

B) If cycle is present in multi-instance RAG then deadlock will always be present. True or False Justify with your own answer with proper example (6 marks)

Q2. A) Three processes on a single processor system share three resource categories, X, Y, and Z. There are 5 units in each resource kind. Consider the following scenario, in which alloc represents the number of units of each resource type allocated to each process, and request indicates the number of units of each resource type requested by a process to complete execution. Which of these procedures will take the LAST amount of time to complete? Make a list of the procedures in a safe order. (10 marks)

	Alloc			Request		
	X	Y	Z	X	Y	Z
P0	1	2	1	1	0	3
P1	2	0	1	0	1	2
P2	2	2	1	1	2	0

B) Variable partition scheme solves the problem of internal fragmentation but not the external fragmentation. Give some comments with proper example. (2.5 marks)

Q3. (A) Suppose we want to synchronize two concurrent processes P and Q using binary semaphores S and T. The code for the processes P and Q is shown below.

```

Process P:
while (1) {
W:
    print '0';
    print '0';
X:
}

Process Q:
while (1) {
Y:
    print '1';
    print '1';
Z:
}

```

Synchronization statements can be inserted only at points W, X, Y and Z. What should be the Binary Semaphore on W, X, Y, Z respectively and what should be the initial value of Binary Semaphore variables S and T in order to get the print output always as 00110011. Explain. (8.5marks)

(B) Consider a system with logical address space of 512 MB. The physical address is 24bits. The memory is byte addressable, the physical address space is divided into 16KB frames then what is the number of pages in the logical address space? (4 marks)

Q.4 (A) How many successive requests of 8KB will be satisfied by using the first fit algorithm assuming variable partition scheme from the Fig.1. (6.5 marks)

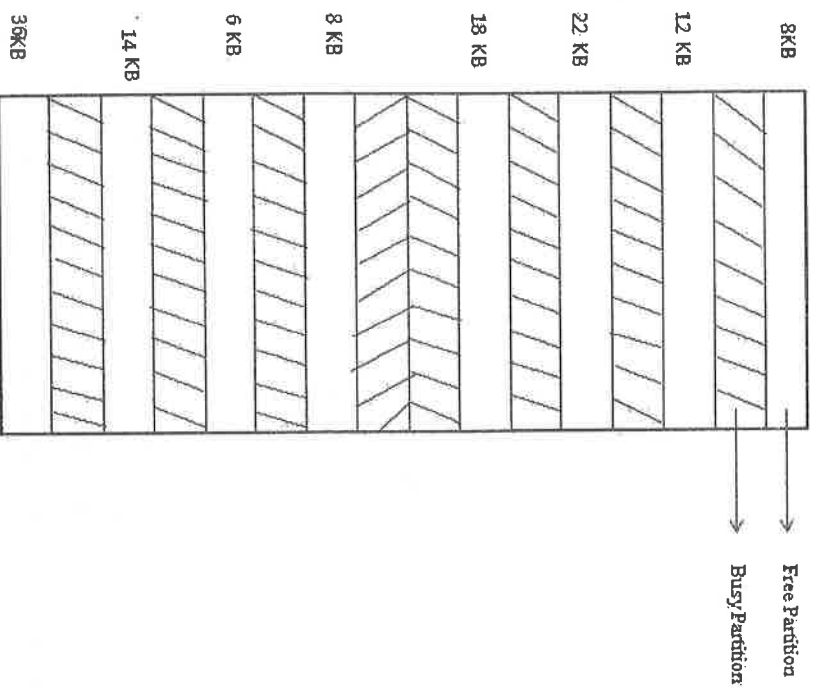


Fig.1

(B) Consider that you work in a software firm as a manager for operating software. How would you go about putting a non-contiguous memory management strategy into practise? Using time and space complexity, compare the first, best, worst, and next fit memory allocations. Also, let us know which complexity, time or space, you will choose and why? (6 marks)

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	12.5
Q2	3,4	CO7, CO5	PO2, PO3, PO4, PO10, PO12	PEO1, PEO3	PSO1, PSO3, PSO4	12.5
Q3	3,4	CO7, CO5	PO2, PO3, PO4, PO10, PO12	PEO1, PEO3	PSO1, PSO3, PSO4	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3, PSO4	12.5

Name of the Examination: CAT 2

School	SCOPE		
Course Code	CSE2008	Course Title	Operating Systems
Class ID	AP2021225000213		
Semester	WIN 2021-22	Date of exam	20.4.2022
Slot	C2+TC2	Faculty Name & ID	Dr. Sanjit Ningthoujam,70147
Duration	90 mins	Total Marks	50

Type your questions here

Q1. A) Consider a system with 3 processes and each process requires 2 resources to complete their execution then what is the minimum number of resources require to ensure deadlock free operation. Explain with the example. (6.5 marks)

B) If cycle is present in single instance RAG then deadlock will always be present. Justify with your own answer with proper example (6 marks)

Q2. A) Three resource categories, X, Y, and Z, are shared by three processes on a single processor system. Each resource type has a total of 5 units. Consider the following scenario, in which the column alloc represents the number of units of each resource type allocated to each process, and the column request represents the number of units of each resource type requested by a process to complete execution. Which of these processes will be the LAST to complete? Write the safe sequence of the processes (10 marks)

	Alloc			Request		
	X	Y	Z	X	Y	Z
P0	1	2	1	1	0	3
P1	2	0	1	0	1	2
P2	2	2	1	1	2	0

B) "Compaction" is good solution for external fragmentation in memory management technique. True or False. Comment on your own words with proper example. (2.5 marks)

Q3. (A) Suppose we want to synchronize two concurrent processes P and Q using binary semaphores S and T. The code for the processes P and Q is shown below.

```

Process P:
while (1) {
W:
    print '0';
    print '0';
X:
}

```

```

Process Q:
while (1) {
Y:
    print '1';
    print '1';
Z:
}

```

Synchronization statements can be inserted only at points W, X, Y and Z. What should be the Binary Semaphore on W, X, Y, Z respectively and what should be the initial value of Binary Semaphore variables S and T in order to get the print output always as 00110011. Explain. (8.5 marks)

(B) Consider a system having a page table with 4K entry and L. A=29 bits then what is the physical address if the system has 512 frames? (4 marks)

Q.4 (A) How many successive requests of 6KB will be satisfied by using the first fit algorithm assuming variable partition scheme from the fig.1. (6.5 marks)

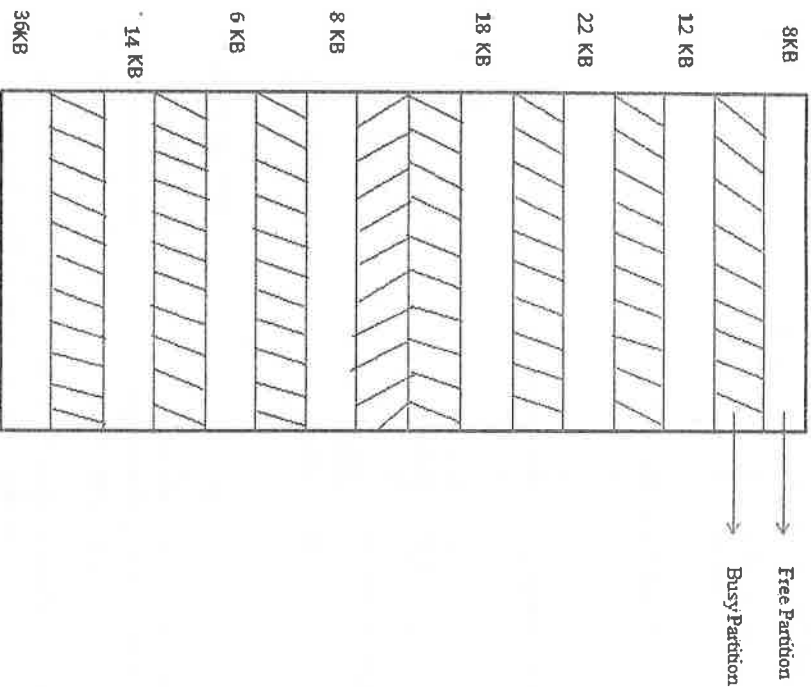


Fig.1

- (B) What hit ratio is required to reduce the EMAT from 300ns without TLB to 250ns with TLB. TLB access time is 60ns. (6 marks)

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	12.5
Q2	3,4	CO7, CO5	PO2, PO3, PO4, PO10, PO12	PEO1, PEO3	PSO1, PSO3, PSO4	12.5
Q3	3,4	CO7, CO5	PO2, PO3, PO4, PO10, PO12	PEO1, PEO3	PSO1, PSO3, PSO4	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3, PSO4	12.5

Name of the Examination: CAT 2

School	SCOPE		
Course Code	CSE2008	Course Title	OPERATING SYSTEMS
Semester	WIN SEM 2-21-22	Date of exam	20-04-2022
Slot	C1+TC1	Faculty Name & ID	P. LALITHA KUMARI
Duration	90 MIN	Total Marks	50

EACH QUESTION CARRIES 12.5 M

Q1.a. Consider the following page reference string:

1, 2, 3, 4, 5, 6, 2, 1, 4, 6, 2, 1, 3, 2, 7, 6, 5, 2, 1, 4, 3, 6

How many page faults would occur for the optimal page replacement algorithm, assuming three frames and all frames are initially empty

b. Given memory partitions of 525 KB, 125 KB, 225 KB, 325 KB, and 625 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 214 KB, 429 KB, 115 KB and 420 KB (in order)? Which algorithm makes the most efficient use of memory?

Q2. a. Consider a logical address space of 50 pages with a 4-KB page size, mapped onto a physical memory of 64 frames.

i. How many bits are required in the logical address?

ii. How many bits are required in the physical address?

b. Consider a computer system with a 64-bit logical address and 4-KB page size. The system supports up to 256 MB of physical memory. How many entries are there in each of the following?

Q3. Consider a system consisting of m resources of the same type being shared by n processes. A process can request or release only one resource at a time. Show that the system is deadlock free if the following two conditions hold:

- The maximum need of each process is between one resource and m resources.
- The sum of all maximum needs is less than $m + n$.

Q4 Consider the following snapshot of a system:

Processes	Allocation ABCD	Max ABCD	Available ABCD
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 2 2	0 6 5 2	
P4	0 0 1 4	0 5 4 5	

Answer the following questions using the banker's algorithm:

- What is the content of the matrix Need?
- Is the system in a safe state?
- If a request from process P1 arrives for (0,3,1,0), can the request be granted immediately?

QP Mapping

Q.No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	4	7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	10M
Q2	4	7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	10M
Q3	4	7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	10M
Q4	3	5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3	10M
Q5	3	5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3	10M

Name of the Examination: CAT 2

School	SCOPE		
Course Code	CSE2008	Course Title	OPERATING SYSTEMS
Semester	WIN SEM 2-21-22	Date of exam	20-04-2022
Slot	C2+TC2	Faculty Name & ID	P. LALITHA KUMARI
Duration	90 MIN	Total Marks	50

EACH QUESTION CARRIES 12.5 M

Q1. Consider the following page reference string 2, 3, 1, 5, 4, 3, 2, 6, 7, 3, 2, 3, 4, 1, 7, 2, 3, 4, 2, 5, 6, 7. Calculate the number of page faults with LRU, FIFO and optimal page replacement algorithms with frame size of 3.

Q2. Consider the following and find out the possible resource allocation sequence with the help of deadlock detection algorithm
processes: p0, p1, p2, p3, p4,
Resources: A, B, C

Allocation: [0 1 0, 2 0 0, 3 0 3, 2 1 1, 0 0 2]

Max: [0 0 0, 2 0 2, 0 0 0, 1 0 0, 0 0 2]

Available: [0 0 1].

Q3. Explain why interrupts are not appropriate for implementing synchronization primitives in multiprocessor systems.

b) Explain the process of converting virtual addresses to physical addresses with a neat diagram.

Q4. Consider the following snapshot of a system:

Processes	Allocation A B C D	Max A B C D	Available A B C D
P0	3 0 0 1	4 2 1 2	3 3 2 1
P1	3 1 2 1	5 2 5 2	
P2	2 1 0 3	2 3 1 6	
P3	1 3 1 2	1 4 2 4	
P4	1 4 3 2	3 6 6 5	

Answer the following questions using the banker's algorithm:

a. Illustrate that the system is in a safe state by demonstrating an order in which the processes may complete.

- b. If a request from process P1 arrives for (1, 1, 0, 0), can the request be granted immediately?
- c. If a request from process P4 arrives for (0, 0, 2, 0), can the request be granted immediately?

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	4	7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	10M
Q2	4	7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	10M
Q3	3	5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3	10M
Q4	4	7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	10M
Q5	3	5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3	10M

Name of the Examination: CAT 2

School	School of Computer Science and Engineering		
Course Code	CSE2008	Course Title	Operating System
Class Number	AP2021225000216	Date of exam	20/04/2022, FN Session
Slot	C1+TC1	Faculty Name & ID	Dr. S.Sudhakar Ilango
Duration	1 hour 30 Minutes	Total Marks	50 Marks

Answer ALL the questions.

Q.No	Question	Marks																																																																									
Q1	Consider the buffer following fixed size buffer. ← Red → ← Green → Red color indicates buffer is full. Green color indicates buffer is empty. i) Justify why Mutex locks are not suitable for this buffer. ii) Illustrate buffer full and empty conditions using semaphore.	12.5 Marks																																																																									
Q2	Alice and Bob are football players. They are playing for two different teams. Both the teams have equal number of goals. Now, the referee called for a penalty corner. At a time either Alice or Bob will get a chance, the referee can call any one of them if they are free and they should not wait for a long time for the penalty corner. Analyze the situation and find the best solution to meet the above three conditions.	12.5 Marks																																																																									
Q3	Consider the following snapshot of a system. <table><thead><tr><th rowspan="2">P. No</th><th colspan="4">Allocation</th><th colspan="4">Max Demand</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr></thead><tbody><tr><td>P0</td><td>1</td><td>1</td><td>2</td><td>3</td><td>1</td><td>2</td><td>2</td><td>3</td><td>5</td><td>3</td></tr><tr><td>P1</td><td>2</td><td>3</td><td>3</td><td>2</td><td>1</td><td>3</td><td>7</td><td>6</td><td>4</td><td>2</td></tr><tr><td>P2</td><td>2</td><td>3</td><td>5</td><td>4</td><td>1</td><td>4</td><td>5</td><td>7</td><td>6</td><td>3</td></tr><tr><td>P3</td><td>2</td><td>1</td><td>3</td><td>2</td><td>0</td><td>5</td><td>4</td><td>7</td><td>3</td><td>1</td></tr><tr><td>P4</td><td>3</td><td>1</td><td>2</td><td>4</td><td>0</td><td>4</td><td>3</td><td>3</td><td>6</td><td>1</td></tr></tbody></table> i) Illustrate that the system is in safe state by demonstrating an order in which the processes may complete. a. Available (1,2,1,2,1) b. Available (3,1,1,2,1) ii) P1 requests (0,1,0,1,0) after time T₁, check the resource can be granted immediately or not. (Use the available (1,2,1,2,1))	P. No	Allocation				Max Demand				A	B	C	D	A	B	C	D	E	P0	1	1	2	3	1	2	2	3	5	3	P1	2	3	3	2	1	3	7	6	4	2	P2	2	3	5	4	1	4	5	7	6	3	P3	2	1	3	2	0	5	4	7	3	1	P4	3	1	2	4	0	4	3	3	6	1	12.5 Marks
P. No	Allocation				Max Demand																																																																						
	A	B	C	D	A	B	C	D	E																																																																		
P0	1	1	2	3	1	2	2	3	5	3																																																																	
P1	2	3	3	2	1	3	7	6	4	2																																																																	
P2	2	3	5	4	1	4	5	7	6	3																																																																	
P3	2	1	3	2	0	5	4	7	3	1																																																																	
P4	3	1	2	4	0	4	3	3	6	1																																																																	
Q4 a.	Consider seven memory partitions of size 170 KB, 200 KB, 478 KB, 365 KB, 500 KB, 670 KB and 250 KB. These partitions need to be allocated to seven processes, all the processes are arriving at 0 ms. The process requirement is given below:	8.5 Marks																																																																									

	Process number	Memory Requirement
	P100	178KB
	P200	120KB
	P101	318KB
	P490	425KB
	P310	486KB
	P550	219KB
	P210	598KB
	Perform the allocation of processes using-	
	1. First Fit Algorithm	
	2. Best Fit Algorithm	
	3. Worst Fit Algorithm	
Q4.b	Consider the following segment table.	
	Segment Number	Base Address Limit
	1	1000 300
	2	1500 200
	3	2000 450
	4	3000 500
	5	3800 150
	6	4000 360
	The Logical address is given below find the corresponding physical address.	
	i) [1,299]	
	ii) [2,200]	
	iii) [3,360]	
	iv) [4,500]	
	v) [5,100]	
	vi) [6,350]	
		4 Marks

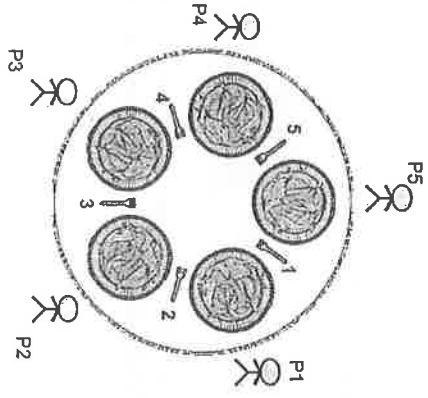
QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO2	PO1, PO2, PO4	PSO1, PSO2	PEO2, PEO3	12.5
Q2	3	CO2	PO1, PO2, PO4	PSO1, PSO2	PEO2, PEO3	12.5
Q3	3	CO7	PO2, PO3, PO4, PO10, PO12	PSO1, PSO3, PSO4	PEO1	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PSO1, PSO2, PSO3, PSO4	PEO3	12.5

Name of the Examination: CAT 2

School	School of Computer Science and Engineering		
Course Code	CSE2008	Course Title	Operating System
Class Number	AP2021225000217	Date of exam	20/04/2022, AN Session
Slot	C2+TC2	Faculty Name & ID	Dr. S.Sudhakar Ilango
Duration	1 hour 30 Minutes	Total Marks	50 Marks

Answer ALL the questions.

Q.No	Question	Marks																																																																														
Q1	Consider the following snapshot of Dining Philosophers.  Illustrate the following scenarios using semaphore. i) Philosopher P3 and P1 is eating. Explain with suitable snapshot. ii) While Philosopher P3 and P1 are eating, Philosopher P2 would like to eat. Explain the blocking condition and when P2 can eat.	12.5 Marks																																																																														
Q2	Kabaddi is basically a combative sport, with seven players on each side; At a time one raider can go for a ride. The raider should tag a member of the opposite team and return back to his half. Currently there are five raiders available named K2,K4,K5,K6 and K7 respectively. Now, K4 is going for the raid, by that time K5 and K6 are also trying to go, but K4 is getting the chance. Analyze the situation and find the feasible solution using process synchronization method. (Use multi-process solution)	12.5 Marks																																																																														
Q3	A system has four processes and five allocatable resources. The current allocation and maximum needs are as follows: <table><tr><th>P</th><th colspan="5">Allocated</th><th colspan="5">Maximum</th><th colspan="3">Available</th></tr><tr><td>A</td><td>3</td><td>2</td><td>3</td><td>3</td><td>3</td><td>4</td><td>5</td><td>3</td><td>4</td><td>5</td><td>1</td><td>1</td><td>X</td><td>2</td><td>3</td></tr><tr><td>B</td><td>3</td><td>3</td><td>2</td><td>2</td><td>1</td><td>3</td><td>4</td><td>4</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td><td></td></tr><tr><td>C</td><td>2</td><td>2</td><td>2</td><td>2</td><td>1</td><td>3</td><td>3</td><td>5</td><td>3</td><td>3</td><td>3</td><td></td><td></td><td></td><td></td></tr><tr><td>D</td><td>3</td><td>3</td><td>3</td><td>3</td><td>1</td><td>3</td><td>6</td><td>5</td><td>5</td><td>3</td><td>3</td><td></td><td></td><td></td><td></td></tr></table> a) What is the smallest value of X for which this is a safe state?	P	Allocated					Maximum					Available			A	3	2	3	3	3	4	5	3	4	5	1	1	X	2	3	B	3	3	2	2	1	3	4	4	4	3	3					C	2	2	2	2	1	3	3	5	3	3	3					D	3	3	3	3	1	3	6	5	5	3	3					12.5 Marks
P	Allocated					Maximum					Available																																																																					
A	3	2	3	3	3	4	5	3	4	5	1	1	X	2	3																																																																	
B	3	3	2	2	1	3	4	4	4	3	3																																																																					
C	2	2	2	2	1	3	3	5	3	3	3																																																																					
D	3	3	3	3	1	3	6	5	5	3	3																																																																					

b) If a request from process D arrives for (0, 0, 1, 1, 0), then can the request be grant immediately? Take the available as [1,2,4,3,3].Find the new snapshot.

Q4.a Consider the following segment table.

Segment Number	Base Address	Limit
1	500	800
2	1500	200
3	1800	380
4	2300	570
5	3000	698
6	3800	412

The Logical address is given below find the addressing error.

- [1,756]
- [2,199]
- [3,380]
- [4,500]
- [5,100]
- [6,412]

4
Marks

Q4.b Consider the following contiguous memory segments.

225KB	350KB	250KB	100KB	600KB	450KB	650KB	100KB	580KB
-------	-------	-------	-------	-------	-------	-------	-------	-------

Shaded memory segments are already allocated to some processes at time T_0 . These partitions need to be allocated to five processes. The process requirement is given below:

P.No	Arrival Time	Memory requirement
P1	0	240KB
P2	2	328KB
P3	1	395KB
P4	2	578KB
P5	3	640KB
P6	4	450KB

Perform the allocation of processes using-

- First Fit Algorithm
- Best Fit Algorithm
- Worst Fit Algorithm

8.5
Marks

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO2	PO1, PO2, PO4	PSO1, PSO2	PEO2, PEO3	12.5
Q2	3	CO2	PO1, PO2, PO4	PSO1, PSO2	PEO2, PEO3	12.5
Q3	3	CO7	PO2, PO3, PO4, PO10, PO12	PSO1, PSO3, PSO4	PEO1	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PSO1, PSO2, PSO3, PSO4	PEO3	12.5

QUESTION PAPER

Name of the Examination: CAT 2 (Winter 2021-22)

School	SCOPE		
Course ID	AP2021225000218		
Course Code	CSE2008	Course Title	Operating System
Semester	Winter	Date of exam	20-04-2022
Slot	C1 and TC1	Faculty Name & ID	Dr. Nileshchandra Kalbarao Pikle (Emp Id 70225)
Duration	1 hour 30 minutes	Total Marks	50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Given a description of The-Cigarette Smokers Problem

There are four processes in this problem: Four smoker processes and an agent process. Each of the smoker processes will make a cigarette and smoke it. To make a cigarette requires tobacco, paper, glue and matches. Each smoker process has one of the four items. I.e., one process has tobacco, another has paper, third has glue and a fourth has matches. The agent has an infinite supply of all three. The agent places three of the four items on the table, and the smoker that has the fourth item makes the cigarette (i.e. only one smoker is allowed to smoke at a time). Explain in detail how synchronization can be achieved using Semaphores for the aforementioned scenario.

12.5 Marks

Q. 2 Consider two cooperating processes: *Tick* and *Tock* sharing a common variable *score*. The code snippet for processes mentioned below

```
Procedure Tick{  
    score = score + 1; // score variable updation  
}
```



```
Procedure Tock{  
    score = score - 1; // score variable updation  
}  
}
```

What could possibly go wrong if all processes try to execute concurrently with preemption? Provide a correct solution to this critical section problem using semaphore.

12.5 Marks

Q. 3 Assume that there are 5 processes, P0 through P4, and 4 types of resources. At T0 we have the following system state:

Max Instances of Resource Type A = 3 (2 allocated + 1 Available)

Max Instances of Resource Type B = 17 (12 allocated + 5 Available)

Max Instances of Resource Type C = 16 (14 allocated + 2 Available)

Max Instances of Resource Type D = 12 (12 allocated + 0 Available)

Using Banker's Algorithm,

1. calculate the content of the *need* matrix
2. Is the system in a safe state? if yes then mention the safe sequence.
3. Determine the total amount of resources of each type?

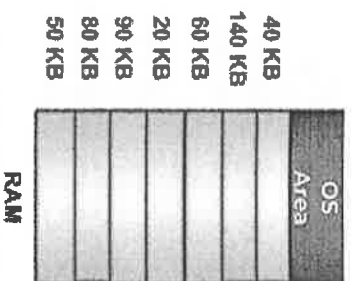
12.5 Marks

Q. 4 Consider main memory is divided into two partitions: user and OS area. User area is further divided into fixed partitions of different sizes as mentioned in the diagram. Five processes of size 30 KB, 80 KB, 140 KB, 20 KB, and 10 KB respectively (assume FCFS order of arrival) are need to be allocated in the main memory, provided that

- a. no two or more processes can be placed in one partition.
- b. a process cannot be placed into two partitions simultaneously.

1. Apply first fit, best fit and worst fit policy to allocate free partitions to the processes and find out total internal fragmentation in each case.

12.5 Marks



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO2	PO1, PO2, PO4	PEO2, PEO3	PSO1, PSO2	12.5
Q2	3	CO2	PO1, PO2, PO4	PEO2, PEO3	PSO1, PSO2	12.5
Q3	3	CO7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO1	PSO1, PSO2, PSO3, PSO4	12.5

QUESTION PAPER

Name of the Examination: CAT 2 (Winter 2021-22)

School	SCOPE		
Course ID	AP2021225000219		
Course Code	CSE2008	Course Title	Operating System
Semester	Winter	Date of exam	20-04-2022
Slot	C2 and TC2	Faculty Name & ID	Dr. Nileschandra Kalbarao Pikle (Emp Id 70225)
Duration	1 hour 30 minutes	Total Marks	50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Given a description of sleeping barber problem

There is a barber shop which has one barber, one barber chair, and n chairs for waiting customers if there are any to sit on the chair.

- If there is no customer, then the barber sleeps in his own chair.
- When a customer arrives, he has to wake up the barber.
- If there are many customers and the barber is cutting a customer's hair, then the remaining customers either wait if there are empty chairs in the waiting room or they leave if no chairs are empty.

12.5 Marks

Q. 2 Assume there are 10 concurrent processes $P_1, P_2, \dots, P_{10}$. All 10 processes have a common critical section. At a time maximum 4 processes are allowed to be entered in the critical section. A counting semaphore variable S is initialized to 4. A following solution to the critical section satisfies mutual exclusion property. However it suffers from a problem of *busy waiting*. Modify the below mentioned code so that busy waiting can be avoided. Explain it by considering the example.

```

Process Pi{
    Wait(S);
Critical Section
Signal(S);
Remainder section
}

```

where Wait() and Signal() operations are atomic and defined as below:

Wait(S):

while($S \leq 0$);

$S = S - 1$;

Signal(S):

$S = S + 1$;

12.5 Marks

Q. 3 Assume that we have the following resources:

- 7 Pen drives (PD)
- 5 Printers (P)
- 7 Scanners (S)

Assume there are four processes. The available resources are already allocated as per the matrix table below.

Process	Allocation			Max			Available		
	PD	P	S	PD	P	S	PD	P	S
P1	1	0	2	6	5	3			
P2	3	1	0	4	2	1			
P3	0	1	1	5	1	2			
P4	1	0	1	1	0	2			
P5	1	0	2	5	2	3			

1. calculate the content of the *need* matrix
2. Calculate available resources
3. Is the system in a safe state? if yes then mention the safe sequence.

12.5 Marks

Q. 4 Consider a main memory divided into two partitions OS area and User area. Total size of memory is 256 KB out of which 40 KB is allocated to the OS area. 5 processes to be executed based on their arrival time (following FCFS technique). The details of the processes are mentioned in the table below.

Process	Memory	CPU Burst	Arrival Time
P1	60 KB	8	0
P2	100 KB	6	1
P3	30 KB	2	5
P4	70 KB	4	7
P5	50 KB	3	9
P6	60 KB	2	11

Assuming the memory allocation strategy as multiprogramming with variable partitioning (MVT),

1. Show status of memory at each time stamp.
2. Calculate total fragmentation each time stamp.

12.5 Marks

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO2	PO1, PO2, PO4	PEO2, PEO3	PSO1, PSO2	12.5
Q2	3	CO2	PO1, PO2, PO4	PEO2, PEO3	PSO1, PSO2	12.5
Q3	3	CO7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO1	PSO1, PSO2, PSO3, PSO4	12.5

QUESTION PAPER

Name of the Examination: CAT 2 (Winter 2021-22)

School	SCOPE		
Course ID	AP2021225000220		
Course Code	CSE2008	Course Title	Operating System
Semester	Winter	Date of exam	23-04-2022
Slot	F2 and TF2	Faculty Name & ID	Dr. Nileschandra Kalbarao Pikle (Emp Id 70225)
Duration	1 hour 30 minutes	Total Marks	50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q. 1 Consider three cooperating processes *Rock*, *Paper* and *Scissor*, sharing a common variable *score*. The code snippet for processes mentioned below

Procedure *Rock*{

score = *score* * 2; // *score* variable updation

}

Procedure *Paper*{

score = *score* + 1; // *score* variable updation

}

Procedure *Scissor*{

score = *score* - 1; // *score* variable updation

}

}

What could possibly go wrong if all processes try to execute concurrently?
Provide a correct solution to this critical section problem using semaphore.

Q. 2 Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared Boolean variables S1 and S2 are randomly assigned.

Method used by P1	Method used by P2
while (S1 == S2); Critical Section S1 = S2	while (S1 != S2); Critical Section S2 = !S1

Which of the following properties can be satisfied? Explain the scenario in detail.

1. Mutual exclusion
2. Progress
3. Busy wait
4. Bounded wait

12.5 Marks

Q. 3 Consider the following snapshot of a system:

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	1	1	2	4	3	3	2	1	0
P1	2	1	2	3	2	2			
P2	4	0	1	9	0	2			
P3	0	2	0	7	5	3			
P4	1	1	2	1	1	2			

1. calculate the content of the *need* matrix?
2. Is the system in a safe state? if yes then mention the safe sequence.
3. Determine the total amount of resources of each type?

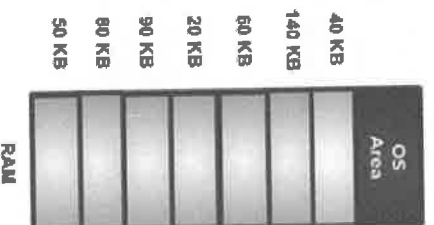
12.5 Marks

Q. 4 Consider main memory is divided into two partitions: user and OS area. User area is further divided into fixed partitions of different sizes as mentioned in the diagram. Five processes of size 50 KB, 80 KB, 40 KB, 20 KB, and 100 KB respectively (assume FCFS order of arrival) are need to be allocated in the main memory, provided that

- no two or more processes can be placed in one partition.
- a process cannot be placed into two partitions simultaneously.

- Apply first fit, best fit and worst fit policy to allocate free partitions to the processes and find out total internal fragmentation in each case.

12.5 Marks



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO2	PO1, PO2, PO4	PEO2, PEO3	PSO1, PSO2	12.5
Q2	3	CO2	PO1, PO2, PO4	PEO2, PEO3	PSO1, PSO2	12.5
Q3	3	CO7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	12.5
Q4	4	CO5	PO2, PO3, PO4, PO7, PO10, PO11	PEO1	PSO1, PSO2, PSO3, PSO4	12.5

Name of the Examination: Re CAT 1

School	SCOPE		
Course Code	CSE2008	Course Title	Operating Systems
Semester	WIN 2021-22	Date of exam	
Slot		Faculty Name & ID	
Duration	90 min	Total Marks	50

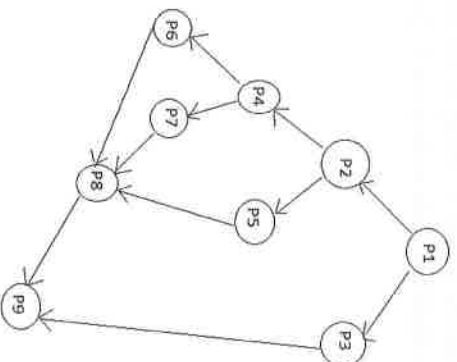
1) How many times “*” will be printed by the following code fragment? Explain. [10M]

```

void main()
{
    if(fork() >= 0)
    {
        printf("*");
        if(fork() == 0)
        {
            printf("*");
        }
        else
        {
            //do nothing
            printf("*");
        }
    }
}

```

2) Implement the below given precedence graph by using fork-join construct. [10M]



3) Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time.

Process	Arrival time	Burst Time	Priority
P1	0	9	2
P2	5	8	0
P3	6	2	3
P4	2	10	1
P5	9	16	4

What is the average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm? Draw the Gantt chart and calculate the waiting times and turn around times of the all processes. [10M]

4) Suppose, there are FIVE concurrent processes P1, P2, P3, P4, and P5 arrive at time 0 with distinct burst times. These processes are going to be executed in a multiprocessing system where THREE processors are available with distinct speeds. The scheduler used by the system is assumed to know the burst times of the processes beforehand. Each process will be assigned to a processor with the below given criteria:

- The fastest processor will be assigned with the process having the longest burst time.
- The slowest processor will be assigned with the process having the shortest burst time.
- Each processor will be assigned with at least one process.

Then, compute the different number of ways that the assignment can be performed. [10M]

5) Consider six processes (P1 to P6), all arriving at time zero, with total execution times in the multiples of 10 milli seconds. Pn process takes $n \times 10$ milliseconds for its execution (P3 takes $3 \times 10 = 30$ milliseconds). These processes are going to be executed in a multiprocessing system where TWO processors are available with same speeds. Each process will be assigned to a processor with the below given criteria:

- Processor 1 will be assigned with a process having the longest burst time.
- Processor 2 will be assigned with a process having the least burst time.

Draw the Gantt charts for Processors 1 and 2. What is the average waiting time of processes. At what time all the processes completes their execution? [10M]

Name of the Examination: RECAT 2

School	SCOPE		
Course Code	CSE2008	Course Title	OPERATING SYSTEMS
Semester	WIN SEM 2-21-22	Date of exam	-05-2022
Slot	C1+TC1	Faculty Name & ID	P. LALITHA KUMARI
Duration	90 MIN	Total Marks	50

EACH QUESTION CARRIES 10M

- 1.Explain Peterson's solution for Critical section problem.
- 2.Illustrate deadlock avoidance with example.
- 3.Explain the paging Techniques.
- 4.Demonstrate page replacement algorithm with example.
- 5.What is dead lock detection and explain how to recover.

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3	10M
Q2	3	5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3	10M
Q3	4	7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	10M
Q4	4	7	PO2, PO3, PO4, PO10, PO12	PEO1	PSO1, PSO3, PSO4	10M
Q5	3	5	PO2, PO3, PO4, PO7, PO10, PO11	PEO3	PSO1, PSO2, PSO3	10M

QUESTION PAPER

Name of the Examination: CAT (Long Summer 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: B+TB+TBB

Date of Exam: 02-07-2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1) Consider the following set of processes. Burst time and arrival time are given in milliseconds. Low value for priority indicates higher priority. Calculate average turnaround time and average waiting time under

- i) Preemptive SJF scheduling algorithm
 - ii) Non-preemptive priority scheduling algorithm
- (12.5M)

Process	Burst Time	Arrival Time	Priority
P1	15	0	3
P2	8	2	1
P3	11	4	4
P4	6	5	2

Q2) A schedule contains two concurrent processes P0, P1 that share two common variables Q, M with initial values 2, 3 respectively. P0 updates Q and M using two instructions S10 and S11, P1 updates the variables Q, M using S20 and S21 instructions. What would be the different values that Q, M can possibly take after the execution of the processes concurrently? (12.5M)

Process P0:	Process P1:
S10: $Q=Q+M$	S20: $Q=Q-1$;
S11: $M=M+Q$;	S21: $M=M+1$;

Q3) Consider 4 processes (P1, P2, P3, P4) and 5 resources (R1, R2, R3, R4, R5). Resource R1 has 2 copies, R2 has 1 copy, R3 has 2 copies, R4 has 3 copies and R5 has 2 copies. Draw a resource allocation graph with the above processes and resources which contain cycles but no deadlock. (12.5M)

Q4) Consider the following snapshot of a system (A, B, C, D are the resources, and P0 to P4 are the processes):

	Max	Allocation	Available
	A B C D	A B C D	A B C D
P0	2 2 0 1	2 1 0 1	1 2 0 1
P1	2 6 4 2	0 0 0 1	
P2	3 4 5 3	2 2 5 1	
P3	5 5 4 1	0 4 2 1	
P4	0 4 5 3	0 2 1 0	

Answer the following questions using the Banker's algorithm:

- What is the content of the matrix Need?
- Is the system in a safe state? If so, write the safe sequence.
- Check whether the request from process P2 for (0, 2, 0, 0) is accepted or not.

(12.5M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	4	2, 5	4	1, 2, 3, 4	12.5
Q2	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12.5
Q3	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12.5
Q4	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12.5

QUESTION PAPER

Name of the Examination: CAT (FALL 2022-2023)

Course Code: CSE2008

Course Title: Operating Systems

Slot: A1+TA1

Date of Exam: 31/10/2022

Duration: 90 Minutes

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. (i) Define spooling. How is it different from buffering? Provide a real-world example in each case. (5 Marks)

(ii) In a multiprogramming system, there are two processes P1 and P2 in the main memory. P1 uses 80% of the time in executing CPU bound instructions and the remaining 20% of its time in performing I/O. Furthermore, P2 uses 90% of the time in executing CPU bound instructions and the remaining 10% of its time in performing I/O. What will be the CPU utilization rate of the system? (5 Marks)

Q2. (i) Consider a multiprogramming system where 4 processes P1, P2, P3, and P4 share the CPU in Round-Robin mode. Let the context switch time be 1 time unit. What is the time quantum 'q' such that every process will get its turn at the CPU in a span of every 10 time units? (5 Marks)

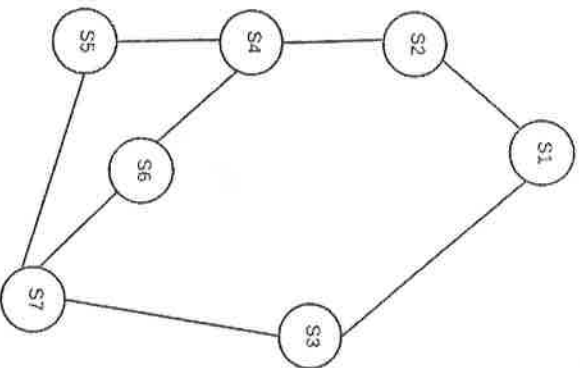
(ii) Suppose that three processes P1, P2, and P3 arrive at time instances 0.0, 0.4, and 1.0 respectively. The burst times of P1, P2, and P3 are 8, 4, and 1 unit(s) respectively. Then, what will be the average turnaround time for these processes if FCFS scheduling algorithm is used? Will the average turnaround time improve if SJF scheduling algorithm is used? (5 Marks)

Q3. (i) Consider the following code snippet:

```
if(fork()) { fork() }  
{  
    fork();  
    fork();  
}
```

How many times "SCOPE" will be printed? Draw the process tree to substantiate the answer. (5 Marks)

(ii) Write a pseudocode using fork and join system calls to implement the below depicted precedence graph. (5 Marks)



- Q4. Consider four concurrent processes P1, P2, P3, and P4 access a shared variable X that is initialized to 200. Processes P1, and P2 add 20, and 30 to the present value of X respectively. Contrastingly, processes P3, and P4 subtract 40, and 50 from the present value of X respectively. The processes are executed on a uniprocessor system with a time-shared OS. Then, what would be the difference between the maximum possible value of X, and the minimum possible value of X after the successful execution of all the four concurrent processes? Assume that the addition and subtraction operations are non-atomic. (10 Marks)

- Q5. (i) Consider a system with three processes that share four instances of a resource type R. The maximum need of each process is k instances of R. Resource instances can be requested and assigned one at a time. Then, what would be the minimum value of k for deadlock to occur?

(5 Marks)

- (ii) A system has 9 instances of a resource type R, which are shared among three concurrent processes P1, P2, and P3. Currently, P1, P2, and P3 are allocated with 3, 1, and 3 instances of R respectively. The maximum requirements of P1, P2, and P3 are 7, 6, and 5 instances of R respectively. Then, can you guarantee that the system is in safe state? If so, then what will be the safe sequence? (5 Marks)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1, 2	1, 2	1, 2	10
Q2	2	1, 2	2, 3	1, 2, 3	2, 3	10
Q3	2	1, 2	2, 3	1, 2, 3	2, 3	10
Q4	3	2, 3	2, 3, 4	2, 3, 4	1, 2, 3, 4	10
Q5	3	2, 3	2, 3, 4	2, 3, 4	1, 2, 3, 4	10

QUESTION PAPER

Name of the Examination: CAT (FALL 2022-2023)

Course Code: CSE2008

Course Title: Operating Systems

Slot: B1+TB1

Date of Exam: 01.11.2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Answer the following questions on the fork. Draw the process hierarchy to represent the process creation. (5M+5M)

a. void main() { int i = 0; for (i = 0; fork();i++){ if(i == 5){ exit();}} How many times fork() is invoked?	b. main(){ print("a\n"); if(fork() == 0){print("b\n"); if(fork() == 0){print("c\n"); exit();} print("d\n");print("e\n");exit();} print("\n");print("g\n");exit();} Mention the process hierarchy and output.
---	---

Q2. Four processes, P100, P200, P300 and P400, are executing, and the below-mentioned table depicts their arrival time and burst time, respectively. (5M+5M)

PID	Arrival Time (in ms)	Burst Time (in ms)
P ₁₀₀	0	3
P ₂₀₀	1	6
P ₃₀₀	4	4
P ₄₀₀	6	2

Find the average waiting time and average turnaround time of the above processes using :

- a. FCFS scheduling
- b. Round Robin scheduling with time quantum = 2 ms (Also depict the ready queue).

Q3.

- a. Five processes, P1, P2, P3, P4 and P5, arrive at the same time at t = 0, with execution times as 5ms, 24ms, 16ms, 10ms and 3ms, respectively. Find the average turn-around time and average waiting time using Shortest Job First (SJF) scheduling algorithm (in ms). (6M)
- b. During the process execution, it was pre-empted due to a timer interrupt. State the logical process of context switching, handling of interrupt requests and fetching of Interrupt Service Routine from Interrupt Vector Table and resumption of the said process again. (4M)

Q4.

- a. Assuming P_x and P_y are two concurrent processes on a time-sharing OS. Which properties of synchronization (Mutual Exclusion, Progress and Bounded Wait) are satisfied? Mention the violations of properties using suitable reasons. Initially, the value of x and y are False. (6M)

$P_x()$	$P_y()$
<pre>while(1){ boolean x = True; while(y); /**Critical Section**/ x = False;} </pre>	<pre>while(1){ boolean y = True; while(x); /**Critical Section**/ y = False;} </pre>

b.

What is the drawback of the "Strict Alteration" technique using the turn variable resolved in Peterson's solution? Explain the significance of the flag variable in Peterson's solution using examples. [4M]

Q5.

Provided five processes P_0, P_1, P_2, P_3, P_4 and four types of resources A, B, C, D . There are in total six instances of resource A , seven instances of resource B , twelve instances of resource C and twelve instances of resource D . The current allocation of resources and maximum requirement is mentioned in the below table. (10M)

Using Banker's algorithm, answer the following questions:

Process ID	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	1	2	X	Y	0	0
P_1	2	0	0	0	2	7	5	0				
P_2	0	0	3	4	6	6	5	6				
P_3	2	3	5	4	4	3	5	6				
P_4	0	3	3	2	0	6	5	2				
Total	4	6	12	12								

- i) Compute X and Y in the available matrix.
- ii) Find if the system is in a safe state? Find the need matrix and the safe sequence.
- iii) P_3 is requesting (0,1,0,0). Check whether it can be granted immediately or not. If yes, get the new snapshot and depict whether the state is safe or not.

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1, 2	1, 2	1, 2	10
Q2	2	1, 2	2, 3	1, 2, 3	2, 3	10
Q3	2	1, 2	2, 3	1, 2, 3	2, 3	10
Q4	3	2, 3	2, 3, 4	2, 3, 4	1, 2, 3, 4	10
Q5	3	2, 3	2, 3, 4	2, 3, 4	1, 2, 3, 4	10

QUESTION PAPER

Name of the Examination: CAT (FALL 2022-2023)

Course Code: CSE2008

Course Title: Operating Systems

Slot: B2+TB2

Date of Exam: 01/11/2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. (a) What is the difference between Multiprogramming and Multitasking operating system? (5 M)
Explain it in detail.

(b) Consider the following code snippet. Explain step-by-step how new processes are created using a process tree diagram. How many times VIT-AP will be printed on execution? (5 M)

```
#include <stdio.h>

#include <sys/types.h>
#include <unistd.h>

int main()
{ int i;
  for(i=1;i<=3;i++){
    fork();
  }
  printf("VIT-AP \n");
  return 0; }
```

Q2. (a) Draw the Process state diagram and explain each state of the process. What are the different types of schedulers and describe their roles in the process management. (6 M)

(b) Differentiate between multilevel queues and multilevel feedback queues. (4 M)

Q3. (a) Consider the following processes scenario. Find out the average turnaround time and average waiting time using Shortest Remaining Time First algorithm (5 M)

Process	Arrival Time	Burst Time
P1	0	7
P2	2	4
P3	4	2
P4	7	1

(b) Consider the following processes scenario. Find out the average turnaround time and average

waiting time using the Preemptive Priority scheduling algorithm.

(5 M)

Process	Arrival Time	Priority	Burst Time
P1	0	3	4
P2	1	2	2
P3	2	4 (Highest)	3
P4	4	1 (Lowest)	2

Note: Higher integer of priority represents high priority.

Q4. Write Peterson Algorithm for two-process synchronization problem and discuss how it satisfies three criteria such as mutual exclusion, progress, and bounded waiting for the solution of critical section problem. (10M)

Q5. Given 5 processes P0, P1, P2, P3, and P4 and 4 types of resources A, B, C, D. There are 9 instances of resource A, 13 instances of resource B, 10 instances of resource C, and 11 instances of resource D available. The current allocation of resources and maximum requirement are mentioned in the below table. Calculate the content of the need matrix. Is the system in a safe state? If yes, then mention the safe sequence using Bankers' Algorithm. (10M)

PID	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	3	2	1	1
P1	2	7	5	0	1	1	0	0				
P2	2	3	5	6	1	2	5	4				
P3	1	6	5	3	0	6	3	3				
P4	1	6	5	6	0	2	1	2				

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	CO1, CO2, CO4	PO1, PO2, PO4	PSO1, PSO2	PEO1, PEO2, PEO3	10
Q2	2	CO3	PO1, PO2, PO3	PSO1, PSO2	PEO1, PEO3	10
Q3	2	CO4	PO2, PO5	PSO1, PSO2, PSO3, PSO4	PEO4	10
Q4	3	CO2	PO1, PO2, PO4	PSO1, PSO2	PEO2, PEO3	10
Q5	3	CO7	PO2, PO3,	PEO1	PSO1, PSO3	10

QUESTION PAPER

Name of the Examination: CAT (FALL 2022-2023)

Course Code: CSE2008

Course Title: Operating Systems

Slot: C1+TC1

Date of Exam: 02/11/2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. a) Explain the functions of Kernel and Shell of an operating system. (5M)

b) A process executes the following code. What is the output of the executed code? Explain step-by-step how outputs are generated using a process tree diagram?

```
void forkexample()
{
    int x = 1;
    if (fork() == 0)
        printf("Child has x = %d\n", ++x);
    else
        printf("Parent has x = %d\n", --x);
}
int main()
{
    forkexample();
    return 0;
}
```

(5M)

Q2. Assume you have the following jobs shown in the table to execute with one processor. Calculate the average waiting time and average turnaround time for the following algorithms. (i) Shortest Remaining Time First (SRTF) (ii) Priority Scheduling (lowest number denotes highest priority). **(10M)**

Process	Arrival Time (ms)	Burst Time (ms)	Priority
P ₀	0	8	5
P ₁	1	4	2
P ₂	2	1	1
P ₃	3	5	3

P ₄	4	4	4
----------------	---	---	---

Q3. (a) What is the role of Process Control Block (PCB) in process context switching?
Explain with a suitable example. (6M)

(b) What are the functions of dispatcher and schedulers? (4M)

Q4. (a) Consider a banking system that maintains an account balance with two functions: withdraw(amt) and deposit(amt). These two functions are passed the amount that is to be deposited or withdrawn from the bank account balance. Assume that a husband and wife share a bank account. Concurrently, the husband calls the deposit() function and the wife calls withdraw(). Describe how a race condition is possible with proper example. (5M)

(b) Explain which of the requirements of synchronization mechanism (Mutual exclusion, Progress and Bounded Wait) are satisfied for the given lock variable synchronization solution. Mention the violations of properties using suitable reasons. (5M)

Entry section:
while(lock != 0);
Lock = 1;
//critical section
Exit section:
Lock = 0;

Q5. Consider the following snapshot of a system

Process	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

(3+4+3 M)

Answer the following questions using Banker's algorithm

- What is the content of "Need" matrix?
- Is the system in a safe state? Justify your answer.
- If a request from P1 arrive for (0 4 2 0), can the request be granted immediately

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1, 2	1, 2	1, 2	10
Q2	2	1, 2	2, 3	1, 2, 3	2, 3	10
Q3	2	1, 2	2, 3	1, 2, 3	2, 3	10
Q4	3	2, 3	2, 3, 4	2, 3, 4	1, 2, 3, 4	10
Q5	3	2, 3	2, 3, 4	2, 3, 4	1, 2, 3, 4	10

QUESTION PAPER

Name of the Examination: CAT (FALL 2022-2023)

Course Code: CSE2008

Course Title: Operating Systems

Slot: D2+TD2

Date of Exam: 03-11-2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1) Illustrate various functionalities of an Operating System.

(10M)

Q2) Illustrate Process State Transition diagram with a neat sketch.

(10M)

Q3) Consider the following set of processes. Burst time and arrival time are given in milliseconds. Low value for priority indicates higher priority. Calculate average turnaround time and average waiting time under

- i) Preemptive SJF scheduling algorithm
- ii) Non-preemptive priority scheduling algorithm

(5M+5M)

Process	Burst Time	Arrival Time	Priority
P1	15	0	3
P2	8	2	1
P3	11	4	4
P4	6	5	2

Q4) Define a semaphore. Illustrate wait and signal semaphore operations. Illustrate semaphore solution for the Critical Section problem. (2M+4M+4M)

Q5) Consider the following snapshot of a system (A, B, C, D are the resources, and P0 to P4 are the processes):

	Max	Allocation	Available
	A B C D	A B C D	A B C D
P0	2 2 0 1	2 1 0 1	1 2 0 1
P1	2 6 4 2	0 0 0 1	
P2	3 4 5 3	2 2 5 1	
P3	5 5 4 1	0 4 2 1	
P4	0 4 5 3	0 2 1 0	

Answer the following questions using the Banker's algorithm for deadlock avoidance:

- Is the system in a safe state? If so, write the safe sequence.
- Check whether the request from process P2 for (0, 2, 0, 0) is accepted or not.

(5M+5M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1, 2	1	1	10
Q2	2	4	2, 5	4	1, 2, 3, 4	10
Q3	2	4	2, 5	4	1, 2, 3, 4	10
Q4	3	3	1, 2, 3, 5	1, 3	1, 2, 4	10
Q5	3	3	1, 2, 3, 5	1, 3	1, 2, 4	10

QUESTION PAPER

Name of the Examination: CAT (FALL 2022-2023)

Course Code: CSE2008

Slot: E2+TE2

Duration: 90 min

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1.

- a) Discuss time sharing operating system and distributed operating system with their advantages and disadvantages. [5]
- b) Discuss how child process can be created using fork() system call. Draw the process hierarchy to represent the process creation with an example. [5]

Q2.

- a) Describe how Scheduler dispatch works and how process gets interrupt with the help of process state diagram. [5]
- b) Explain the Process Control block in details. [5]

Q3. Consider the following process table.

Process ID	Arrival Time (ms)	Burst Time (ms)
P0	1	4
P1	2	2
P2	0	5
P3	1	8
P4	2	6

- a) Find the average waiting time and average turnaround time of all the process using shortest remaining time first scheduling (SRTF) pre-emptive algorithm. [5]
- b) Find the average waiting time and average turnaround time using round robin scheduling algorithm with time quantum 2 ms. [5]

Q4. Explain Peterson's Solution to critical section in details.

[10]

Q5. Consider the following system with five processes <P1, P2, P3, P4, P5> and four resources <R1, R2, R3, R4>. The following table shows the snapshot of the system at time = t_0 . [10]

Process	Allocated				Maximum				Available			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
P1	0	0	1	2	0	0	1	2	1	5	2	0
P2	1	0	0	0	1	7	5	0				
P3	1	3	5	4	2	3	5	6				
P4	0	6	3	2	0	6	5	2				
P5	0	0	1	4	0	6	5	6				

Determine the state is safe or not using Banker's algorithm. If safe, then calculate the safe sequence.
If P2 requests <0,4,2,0> resources after the Banker's algorithm implementation can it be granted?

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	2	PO1, PO2	PEO1	PSO1	10M
Q2	2	2	PO1, PO2	PEO1	PSO1,2	10M
Q3	2	2	PO2, PO3	PEO1, PEO2, PEO3, PEO4	PSO2,3	10M
Q4	3	3	PO1, PO2, PO3, PO4	PEO1, PEO3	PSO1, PSO2, PSO4	10M
Q5	3	2, 3	PO2, 3, PO4	PEO4	PSO1, PSO2, PSO3, PSO4	10M

QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: C1+TC1

Date of Exam: 25-05-2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Answer ALL the questions.
2. Any assumptions made should be clearly stated.

Q1.

Consider the following process table. Each process assigned a numerical priority, with a higher number indicating a higher relative priority. The length of a time quantum is 5 ms. If a process is pre-empted by a higher-priority process, the pre-empted process is placed at the end of the queue.

Thread	Priority	Burst Time	Arrival Time
P100	25	20	0
P101	32	14	5
P110	41	8	5
P125	28	5	10
P130	51	19	10
P136	10	8	15

Calculate the average waiting time and average turnaround time for the following scheduling algorithms.

- i) Round robin scheduling (6 Marks)
- ii) Priority Scheduling (Non-Preemptive) (6 Marks)

Q2.

During the summer holidays, Arun, Surya, and Vijay went on a very exciting exploration of the treasure hunt. They have gone to a nearby forest and they found a treasure inside the Tiger cave. If the tiger is available inside the cave, then they cannot take the treasure. If the tiger is not available, then they can go either individually or in a group to take the treasure. In the cave, only one tiger can stay. Find a feasible solution to solve this (synchronisation) problem using conditional variables. (12 marks)

Q3.

In a virtual memory system, size of virtual address is 64-bit, size of physical address is 60-bit, page size is 8 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. (12 Marks)

- i) Calculate the Main memory size.
- ii) Calculate the number of frames in the in main memory.
- iii) Calculate the page table size
- iv) Find the number of bits used for offset.

Q4.a

Consider the following reference string.
a,b,c,a,e,c,d,e,f,a,b,c,e,a,b,d,c,e,f,b

Check Belady's anomaly in page replacement with 3,4 frames. Give your findings. (6 Marks)

Q4.b Consider the file storage system has 24 blocks. The file block allocations are given below:

Directory 1		
File name	Start	Length
sample.txt	3	2
fifo.c	6	3

Directory 2			
File name	Start	end	
lru.c	10	19	10->15->16->19
Test.java	12	2	12->11->2

Directory 3		
File name	Indexed block	Blocks allocated
pipe.c	20	21,22,0

Show the following file allocation with suitable diagram. (6 Marks)

- Contiguous allocation (Use Directory 1)
- Linked allocation (Use Directory 2)
- Indexed allocation (Use Directory 3)

Q5 Consider a disk queue with requests for I/O to blocks on cylinders 135,46,176,86,98,16,53,60

Consider a disk with 200 cylinders. The head is currently at cylinder number 75. Find the total head movement (in number of cylinders) incurred while servicing these requests using the following algorithms. Head moves towards the smallest cylinder number. (12 marks)

- SSTF
- SCAN
- LOOK
- C-SCAN

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	4	2, 5	4	1, 2, 3, 4	12
Q2	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12
Q3	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	12
Q4.a	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	6
Q4.b	5	6	2, 5, 8, 10, 11	3, 4	1, 2	6
Q5	6	6	2, 5, 8, 10, 11	3, 4	1, 2	12

QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: C2+TC2

Date of Exam: 25-05-2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Answer ALL the questions.
2. Any assumptions made should be clearly stated.

Q1.

Consider the below processes with their burst time and arrival time. Draw a Gantt chart and completion time and calculate completion time, average waiting time, and average turnaround time using Shortest Remaining Time First and Priority Scheduling (Preemptive mode) algorithms. The largest number has low priority. (12 Marks)

Process ID	Arrival Time	Priority	Burst Time
P100	0	15	28
P110	15	8	11
P105	10	3	10
P150	15	7	15
P155	19	4	18
P160	23	9	14

Q2.

Given 5 processes P0, P1, P2, P3, P4 and 4 types of resources A, B, C, D. There are 9 instances of resource A, 13 instances of resource B, 10 instances of resource C and 11 instances of resource D available. The current allocation of resources and maximum requirement is mentioned in the below table. (12 Marks)

Process ID	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	3	2	1	1
P1	2	7	5	0	1	1	0	0				
P2	2	3	5	6	1	2	5	4				
P3	1	6	5	3	0	6	3	3				
P4	1	6	5	6	0	2	1	2				

Using Banker's algorithm, answer the following questions:

- i) What are the contents of the Need matrix?
- ii) Find if the system is in safe state? If it is, find the safe sequence.
- iii) P1 is requesting (1,1,1,1), check whether it can be granted immediately or not. If yes, get the new snapshot.

Q3.

A certain computer system has the paging architecture for virtual memory. Physical Address Space consists of 4GB and Virtual Address Space consists of 64 GB. A page size is equal to 8KB. Page table entry size is 4 bytes. Considering this system configuration and find the followings: (12 Marks)

- Total Number of Pages
- Total Number of Frames
- Number of bits required to reference page offset
- Number of bits required to reference a frame
- Number of bits in physical address
- Size of page table

Q4.a

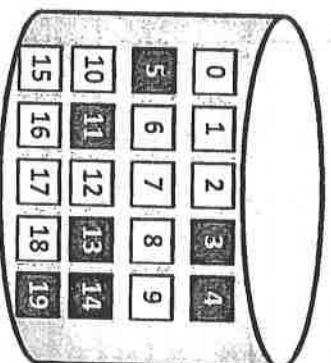
Consider the following page reference.

x,y,z,w,u,v,x,y,u,v,w,z,y,x,w,y,u,v,w,z

Find the page fault using Least recently used page replacement and Optimal page replacement algorithms with necessary snapshot. Use 3 frames. (6 Marks)

Q4.b

Consider a computer HDD has 20 sectors, where each sector size is 2KB. There are 3 files file A, File B and File C with size 4KB, 8KB and 12KB respectively. The current status of the disk sector is mentioned with sector numbers in the below diagram. White sectors indicate FREE and Black sectors indicates USED. (6 Marks)



- First allocate memory to File C using contiguous memory allocation scheme.
- Allocate memory to File B using indexed memory allocation.
- Allocate memory to File C using linked list allocation.

Q5

Consider a disk with 200 cylinders numbered from 0 to 199 and a queue with requests for 8 cylinders as mentioned below

87, 160, 40, 140, 36, 72, 66, 15

Current position of the read/write head is at cylinder number 60. Calculate average read/write head movements using C-SCAN algorithm, LOOK algorithm, FCFS algorithm and SSTF algorithm. Assume first direction of read/write head is towards largest track number in the queue. (12 Marks)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	4	2, 5	4	1, 2, 3, 4	12
Q2	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12
Q3	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	12
Q4.a	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	6
Q4.b	5	6	2, 5, 8, 10, 11	3, 4	1, 2	6
Q5	6	6	2, 5, 8, 10, 11	3, 4	1, 2	12

TEMPLATE FOR QUESTION PAPER

Name of the Examination: FAT

School	School of Computer Science and Engineering			
Course Code	CSE2008	Course Title	Operating System	
Class Number	AP2021225000208	Date of exam	27/05/2022	
Slot	E1+TE1	Faculty Name & ID	Dr. Sushil Kumar Tiwari (70287)-	
Duration	3 hours	Total Marks	60	

1. Consider the following table of arrival time and burst time for three processes P0, P1 and P2.

Process	Arrival time	Burst Time
P0	0 ms	9 ms
P1	1 ms	4 ms
P2	2 ms	9 ms

Apply the pre-emptive and non-pre-emptive shortest job first scheduling algorithm and compare the both approaches on the basis of following parameters. [12]

- Average turnaround time
- Average waiting time
- Average response time

2. A system has 4 processes and 5 allocatable resource. The current allocation and maximum needs are given in the following table. If Available = [0 0 X 1 1], what is the smallest value of x for which this is a safe state? [12]

Processes	Allocated					Maximum				
A	1	0	2	1	1	1	1	2	1	3
B	2	0	1	1	0	2	2	2	1	0
C	1	1	0	1	1	2	1	3	1	1
D	1	1	1	1	0	1	1	2	2	0

3. (a) Consider a machine with 64 MB physical memory and a 32-bit virtual address space. If the page size is 4KB, what is the approximate size of the page table? [6]
- (b) How does paging differ from the Inverted paging? In what circumstances, utilization of inverted paging would be more appropriate in comparison to paging. If you do so, what problems will arise? [6]
4. Consider a reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. The number of frames in the memory is 3. Find out the number of page hit and page faults ratio with respect to following page replacement techniques. Also, determine, if the number of frames are increased by one frame, is there a possibility to decrease the page fault ratio? If yes, then which of the following page replacement policy will suffer from this issue. Justify your answer. [12]
- Optimal Page Replacement Algorithm
 - FIFO Page Replacement Algorithm
 - LRU Page Replacement Algorithm
5. Consider a disk pack with 32 surfaces, 256 tracks per surface, 512 sectors per track and 1024 bytes per sector. Then, answer the following questions. [12]
- If the format overhead is 32 bytes per sector, what is the formatted disk space?
 - If the diameter of innermost track is 21 cm with 2 KB/cm, what is the capacity of one track?
 - If the disk is rotating at 3600 RPM, what is the data transfer rate?

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	CO3	PO3, PO5			12
Q2	3	CO7	PO4, PO10			12
Q3	4	CO5	PO7, PO10			12
Q4	5	CO5	PO7, PO10			12
Q5	6	CO5	PO7, PO10			12

QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE2008

Course Title: Operating System

Slot: F2+TF2

Date of Exam: 28.05.2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Consider the following set of processes with CPU burst time, arrival time and priority given in milliseconds. (Low priority number has high response). **(12M)**

Process	Arrival Time	Burst Time	Priority
P1	12	16	12
P2	9	15	25
P3	4	6	4
P4	0	10	20
P5	20	2	16
P6	18	18	12
P7	10	7	15

Draw Gantt charts illustrating the execution of these processes. Calculate waiting time of each process and average waiting time. Calculate turnaround time of each process and average turnaround time for the following scheduling algorithms.

1. SJF (Preemptive and Non-Preemptive)
2. Priority scheduling.

Q2. Assume that there are seven processes, P_0 through P_6 , and four types of resources. At T_0 we have the following system state: **(12M)**

P. No.	Allocation				Max Demand			
	A	B	C	D	A	B	C	D
P0	3	1	0	0	5	3	1	0
P1	2	1	0	0	2	2	0	1
P2	1	1	1	0	3	3	2	2
P3	3	1	0	2	3	1	0	4
P4	0	2	1	1	1	3	2	2
P5	0	1	1	1	0	5	6	1
P6	1	1	0	2	2	3	2	6

- i) Using Banker's algorithm, calculate the need matrix and illustrate that the system is in safe state by demonstrating an order in which the processes may complete.
- a. Available (2,3,1,1)

If a new request from process P4 arrives for additional resources of (0,1,0,1), can the Banker's algorithm grant the request immediately? Present the new system state using Banker's algorithm.

Q3. Consider the following CPU's reference string: (12M)

0,1,4,7,2,4,5,2,5,3,0,3,1,2,5,6,1,0

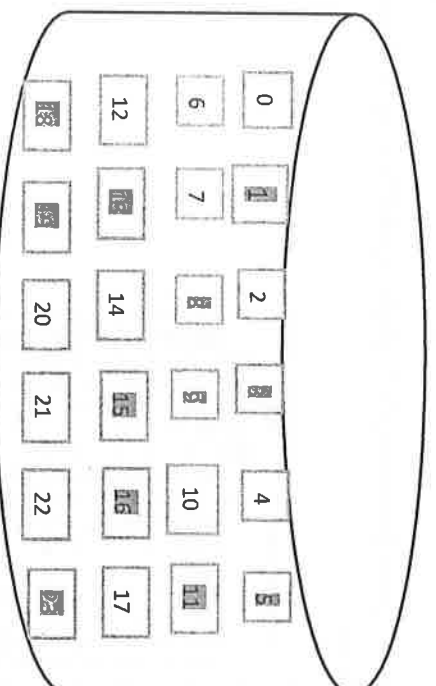
The system has four frames. How many page faults will occur for FIFO, LRU and Optimal algorithm? Find the hit and miss ratio of each algorithm. Find the algorithm which gives efficient page replacement.

Q4. Which of the following logical address will produce trap addressing error? Calculate the physical address if no trap is produced. (6M)

Segment No.	Base	Length
0	131478	900
1	2500	15
2	900	150
3	1110	520
4	280	210
5	1900	668
6	2105	90

a) 0, 900 b) 0,890 c) 1,14 d) 1,9 e) 2,100 f) 3,550 g) 4,200 h) 5,669 i) 6,10

Q4.b. Consider a computer HDD has 24 sectors, where each sector size is 3KB. There are 3 files, File A, File B and File C with size 6KB, 9KB and 18KB respectively. The current status of the disk sector is mentioned with sector numbers in the below diagram. White sectors indicate FREE and shaded sectors indicate USED. (6M)



- i) First allocate memory to File B using contiguous memory allocation scheme.
- ii) Allocate memory to File A using indexed memory allocation.
- iii) Allocate memory to File C using linked list allocation.

Q5. Suppose that a disk drive has 500 cylinders, numbered 0 to 499. The drive is currently serving a request at cylinder 165, and the previous request was at cylinder 167. The queue of pending requests, in FIFO order, is:

(12M)

45,77,155,361,99,380,150,461,470,420,250,410

Starting from the current head position (165), what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms?

- i) FCFS
- ii) SSTF
- iii) SCAN
- iv) C-SCAN (head movement towards lower cylinder number)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	4	2,5	4	1,2,3,4	12
Q2	3	7	2,3,4,10,12	1	1,3,4	12
Q3	4	5	2,3,4,7,10,11	3	1,2,3,4	12
Q4	4&5	5&6	2,3,4,5,7,8,10,11	3,4	1,2,3,4	12
Q5	6	7	2,3,4,10,12	1	1,3,4	12

QUESTION PAPER

Name of the Examination: FAT (Long Summer 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: B+TB+TB

Date of Exam: 29-07-2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Answer ALL the questions.
2. Any assumptions made should be clearly stated.

Q1. Consider the below processes with their burst time and arrival time. Calculate average turnaround time and average waiting time using Non pre-emptive Shortest Job First and Pre-emptive Priority Scheduling algorithms. The largest number has low priority. **(12M)**

Process	Arrival Time	Priority	Burst Time
P1	0	15	28
P2	15	8	11
P3	10	3	10
P4	15	7	15
P5	19	4	18

Q2. Given 5 processes P0, P1, P2, P3, P4 and 4 types of resources A, B, C, D. There are 9 instances of resource A, 13 instances of resource B, 10 instances of resource C and 11 instances of resource D available. The current allocation of resources and maximum requirement is mentioned in the below table. **(12M)**

Process ID	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	3	2	1	1
P1	2	7	5	0	1	1	0	0				
P2	2	3	5	6	1	2	5	4				
P3	1	6	5	3	0	6	3	3				
P4	1	6	5	6	0	2	1	2				

Using Banker's algorithm, answer the following questions:

- i) What are the contents of the Need matrix?
- ii) Find if the system is in safe state? If it is, find the safe sequence.

- Q3.** a) Assume the memory is divided into five partitions with size 250 KB, 400 KB, 360 KB, 500 KB, and 180 KB (in order). Calculate the total internal fragmentation when the processes with size 315 KB, 221 KB, 135 KB, and 284 KB (in order) are placed with
- First-Fit algorithm
 - Best-Fit algorithm
- (8M)**

b) Consider the following segment table:

Segment	Base	Length
0	620	450
1	1800	60
2	170	74
3	2145	321
4	1450	113

What are the physical addresses for the following logical addresses?

(4M)

- 0, 512
- 1, 47
- 2, 13
- 3, 185

- Q4.** Consider a process with 7 pages. If there are 3 frames in main memory, and the reference string is 2, 4, 1, 7, 1, 4, 2, 6, 1, 6, 2, 1, 6, 2, 3, 2, 6, 1, 5, 1, then how many page faults would occur with the following replacement algorithms? **(12M)**

- LRU replacement
- Optimal replacement

- Q5.** Consider a disk with 200 cylinders numbered from 0 to 199 and a queue with requests for 8 cylinders as mentioned below

87, 160, 40, 140, 36, 72, 66, 15

Current position of the read/write head is at cylinder number 60. Calculate total read/write head movement using FCFS algorithm, SSTF algorithm, C-SCAN algorithm and LOOK algorithm. Assume first direction of read/write head is towards largest cylinder number in the queue. **(12M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	4	2, 5	4	1, 2, 3, 4	12
Q2	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12
Q3.a	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	8
Q3.b	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	4
Q4	5	6	2, 5, 8, 10, 11	3, 4	1, 2	12
Q5	6	6	2, 5, 8, 10, 11	3, 4	1, 2	12

QUESTION PAPER

Name of the Examination: FAT (Long Summer 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: B+TB+TB

Date of Exam: 29-07-2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Answer ALL the questions.
2. Any assumptions made should be clearly stated.

Q1. Consider the below processes with their burst time and arrival time. Calculate average turnaround time and average waiting time using Non pre-emptive Shortest Job First and Pre-emptive Priority Scheduling algorithms. The largest number has low priority. **(12M)**

Process	Arrival Time	Priority	Burst Time
P1	0	15	28
P2	15	8	11
P3	10	3	10
P4	15	7	15
P5	19	4	18

Q2. Given 5 processes P0, P1, P2, P3, P4 and 4 types of resources A, B, C, D. There are 9 instances of resource A, 13 instances of resource B, 10 instances of resource C and 11 instances of resource D available. The current allocation of resources and maximum requirement is mentioned in the below table. **(12M)**

Process ID	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	3	2	1	1
P1	2	7	5	0	1	1	0	0				
P2	2	3	5	6	1	2	5	4				
P3	1	6	5	3	0	6	3	3				
P4	1	6	5	6	0	2	1	2				

Using Banker's algorithm, answer the following questions:

- i) What are the contents of the Need matrix?
- ii) Find if the system is in safe state? If it is, find the safe sequence.

Q3.

- a) Assume the memory is divided into five partitions with size 250 KB, 400 KB, 360 KB, 500 KB, and 180 KB (in order). Calculate the total internal fragmentation when the processes with size 315 KB, 221 KB, 135 KB, and 284 KB (in order) are placed with
- First-Fit algorithm
 - Best-Fit algorithm

(8M)

- b) Consider the following segment table:

Segment	Base	Length
0	620	450
1	1800	60
2	170	74
3	2145	321
4	1450	113

What are the physical addresses for the following logical addresses?

(4M)

- 0, 512
- 1, 47
- 2, 13
- 3, 185

Q4.

Consider a process with 7 pages. If there are 3 frames in main memory, and the reference string is 2, 4, 1, 7, 1, 4, 2, 6, 1, 6, 2, 1, 6, 2, 3, 2, 6, 1, 5, 1, then how many page faults would occur with the following replacement algorithms?

(12M)

- LRU replacement
- Optimal replacement

Q5

Consider a disk with 200 cylinders numbered from 0 to 199 and a queue with requests for 8 cylinders as mentioned below

87, 160, 40, 140, 36, 72, 66, 15

Current position of the read/write head is at cylinder number 60. Calculate total read/write head movement using FCFS algorithm, SSTF algorithm, C-SCAN algorithm and LOOK algorithm. Assume first direction of read/write head is towards largest cylinder number in the queue. (12M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	4	2, 5	4	1, 2, 3, 4	12
Q2	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12
Q3.a	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	8
Q3.b	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	4
Q4	5	6	2, 5, 8, 10, 11	3, 4	1, 2	12
Q5	6	6	2, 5, 8, 10, 11	3, 4	1, 2	12

QUESTION PAPER

Name of the Examination: Re-FAT (Winter 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: C1+TC1

Date of Exam:

Duration: 120 min

Total Marks: 60

Instructions:

1. Answer ALL the questions.
2. Any assumptions made should be clearly stated.

Q1. Consider the following set of processes, with the following Arrival time, CPU Burst times given in milliseconds and Priority (High priority number has lower priority).

Process	Arrival Time	Burst Time	Priority
P1	3	13	2
P2	5	7	5
P3	0	18	1
P4	3	8	3
P5	2	6	8
P6	5	4	4

Draw Gantt charts illustrating the execution of these processes. Calculate waiting time of each process and average waiting time. Calculate turnaround time of each process and average turnaround time for the following scheduling algorithms.

1. SJF (Preemptive) (6 Marks)
2. Priority Scheduling (6 Marks)

Q2. Consider the following snapshot of a system.

P. No	Allocation					Max Demand				
	A	B	C	D	E	A	B	C	D	E
P0	0	0	2	3	0	1	1	3	5	3
P1	2	0	0	0	1	3	8	7	1	3
P2	2	3	5	4	1	3	4	7	7	2
P3	0	6	3	2	0	1	7	7	4	2
P4	0	0	2	4	0	1	7	7	7	2

- i) Illustrate that the system is in safe state by demonstrating an order in which the processes may complete.
Use Available as (0,2,1,1,1) (6 Marks)
- ii) P1 requests (0,1,0,1,0) after time T_1 , check the resource can be granted immediately or not. (Use the available (2,3,3,3,2)). Get the safe sequence for new snapshot, if the resource can be granted immediately. (6 Marks)

Q3. Consider a machine with 128 MB physical memory and a 32 bit virtual address space. If the page size is 8 KB. (12 Marks)

- Find the number of bits in physical memory
- Find the number of frames in the Main memory
- Find the number of bits allocated for offset.
- Find the number of entries in page table

Q4.a Consider the following reference string,

1,2,3,1,4,5,1,2,5,4,6,7,1,2,3,1,2,5,7,4

The system has three frames. Compare the number of page faults for LRU and FIFO. (6 Marks)

Q4.b

Illustrate file allocation methods for the following snapshot.
Consider the file storage system has 20 blocks. The file block allocations are given below:

Directory 1		
File name	Start	Length
test.txt	0	3
sfj.c	5	2

Directory 2		
File name	Indexed block	Blocks allocated
ipc.java	11	9,8,6,13

Directory 3		
File name	Start	end
opr.c	7	18
		10->14->16->19

Show the following file allocation with suitable diagram. (6 marks)

- Contiguous allocation (Use Directory 1)
- Linked allocation (Use Directory 3)
- Indexed allocation (Use Directory 2)

Q5 Consider a disk queue with requests for I/O to blocks on cylinders.

25,38,92,86,65,112,128,132,172.

Consider a disk with 200 cylinders. The head is currently at cylinder number 50. Find the total head movement and the seek time (in number of cylinders) incurred while servicing these requests using the FCFS, SSTF, C-SCAN and C-LOOK algorithms.
Head movement towards larger cylinder number. (12 Marks)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	4	2, 5	4	1, 2, 3, 4	12
Q2	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12
Q3	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	12
Q4.a	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	6
Q4.b	5	6	2, 5, 8, 10, 11	3, 4	1, 2	6
Q5	6	6	2, 5, 8, 10, 11	3, 4	1, 2	12

QUESTION PAPER

Name of the Examination: Re-FAT (Winter 2021-2022)

Course Code: CSE2008

Course Title: Operating Systems

Slot: C2+TC2

Date of Exam:

Duration: 120 min

Total Marks: 60

Instructions:

1. Answer ALL the questions.
2. Any assumptions made should be clearly stated.

Q1. Consider the below 5 processes with their burst time, arrival time and priority. Draw a Gantt chart and calculate completion time, average waiting time, and average turnaround time using Preemptive Priority CPU scheduling algorithm and Round robin scheduling algorithm. Use Time Quantum=3ms. (12 Marks)

Process ID	Arrival Time	Burst Time	Priority
P1	0	4	1
P2	0	3	2
P3	6	7	1
P4	11	4	3
P5	12	2	2

Q2. Let there be three Processes P1, P2, P3, P4, P5 and 4 resources A, B, C, D. The maximum resources requirements of the Processes are shown in the below table. (12 Marks)

Process ID	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P1	3	2	3	2	1	2	3	1	2	1	1	1
P2	2	3	1	4	2	1	0	2				
P3	3	1	5	0	2	0	1	0				
P4	2	3	5	3	1	1	1	1				
P5	3	2	1	2	1	0	0	1				

Using Banker's algorithm, answer the following questions:

- i) What are the contents of the Need matrix?
- ii) Find if the system is in safe state? If it is, find the safe sequence.
- iii) P3 is requesting (1, 0, 1, 0). Check whether it can be granted immediately or not.

Q3.

A certain computer system has the paging architecture for virtual memory. Physical Address Space consists of 4GB and Virtual Address Space consists of 16 GB. A page size is equal to 2KB. Page table entry size is 4 bytes. Considering this system configuration and calculate following items. (12 Marks)

- i) Total Number of Pages
- ii) Total Number of Frames
- iii) Number of bits required to reference page offset
- iv) Number of bits required to reference a frame
- v) Number of bits in physical address
- vi) Size of page table

Q4.

Consider the page reference string: 1, 2, 3, 4, 5, 1, 3, 1, 6, 3, 2, 3, 1, 2, 3, 4, 2, 1, 2, 3. (12 Marks)

- i) Check the effect of Belady's anomaly using 3 and 4 frames.
- ii) Find the page fault and Hit Ratio using LRU. Use 3 frames.

Q5

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The head is initially at cylinder number 53 moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. Calculate the total head movement (in number of tracks) incurred while servicing these requests using (12 Marks)

1. The SSTF scheduling algorithm
2. First Come First Serve algorithm
3. SCAN algorithm
4. C-Look algorithm

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	4	2, 5	4	1, 2, 3, 4	12
Q2	3	3	1, 2, 3, 5	1, 3	1, 2, 4	12
Q3	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	12
Q4.	4	5	2, 3, 4, 7, 10, 11	3	1, 2, 3, 4	12
Q5	6	6	2, 5, 8, 10, 11	3, 4	1, 2	12